

➤ Anaconda

- Anaconda is a package manager, an environment manager, and Python distribution that contains a collection of many open source packages.
- Anaconda is an open-source software that contains Jupyter, spyder, etc that are used for large data processing, data analytics, heavy scientific computing. Anaconda works for R and python programming language.

➤ Installing Anaconda on Windows

- Head over to anaconda.com and install the latest version of Anaconda. Make sure to download the “Python 3.7 Version” for the appropriate architecture.

Windows | macOS | Linux

Anaconda 2019.10 for Windows Installer

Python 3.7 version

Download

64-Bit Graphical Installer (462 MB)
32-Bit Graphical Installer (410 MB)

Python 2.7 version

Download

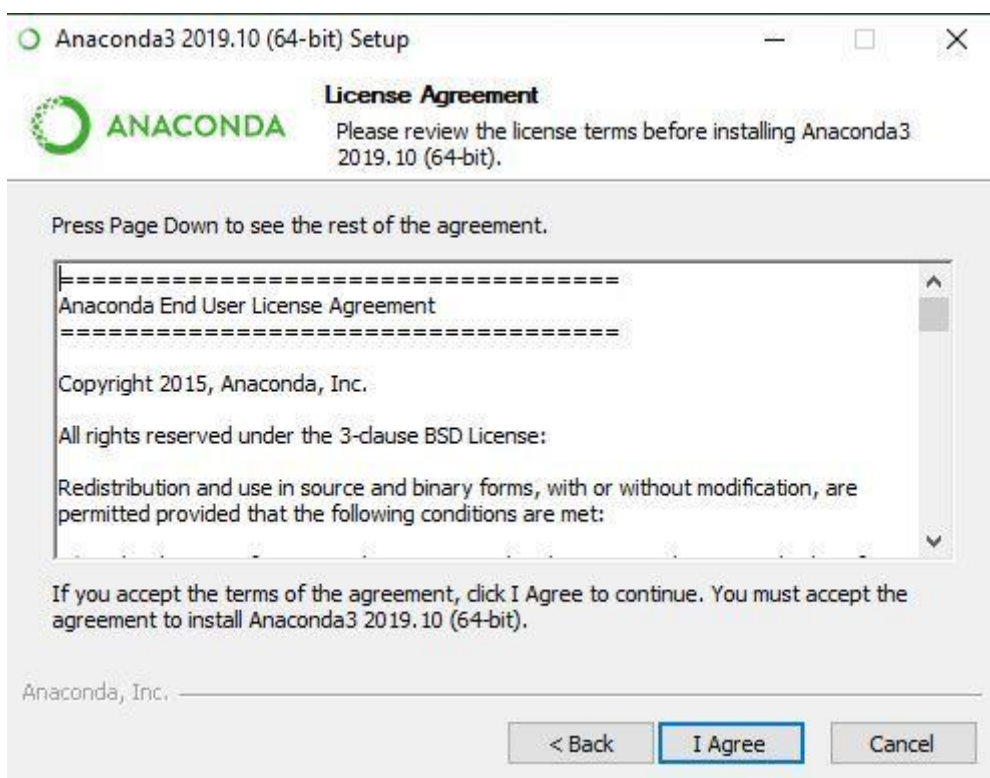
64-Bit Graphical Installer (413 MB)
32-Bit Graphical Installer (356 MB)

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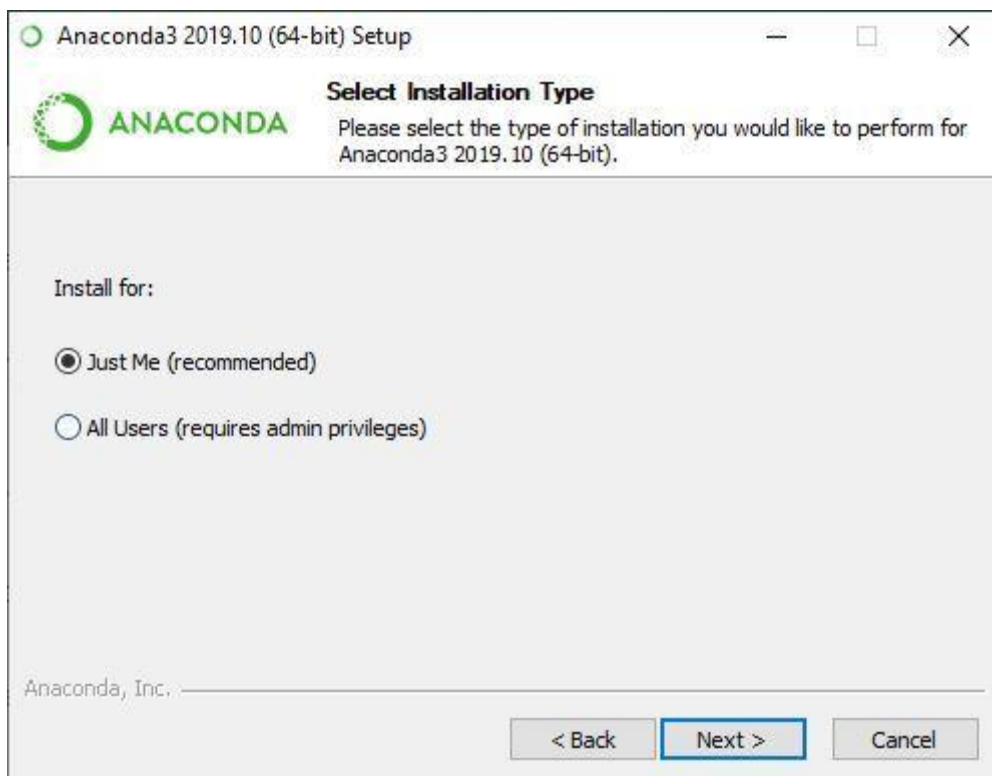
- **Begin with the installation process:**
- Getting Started:



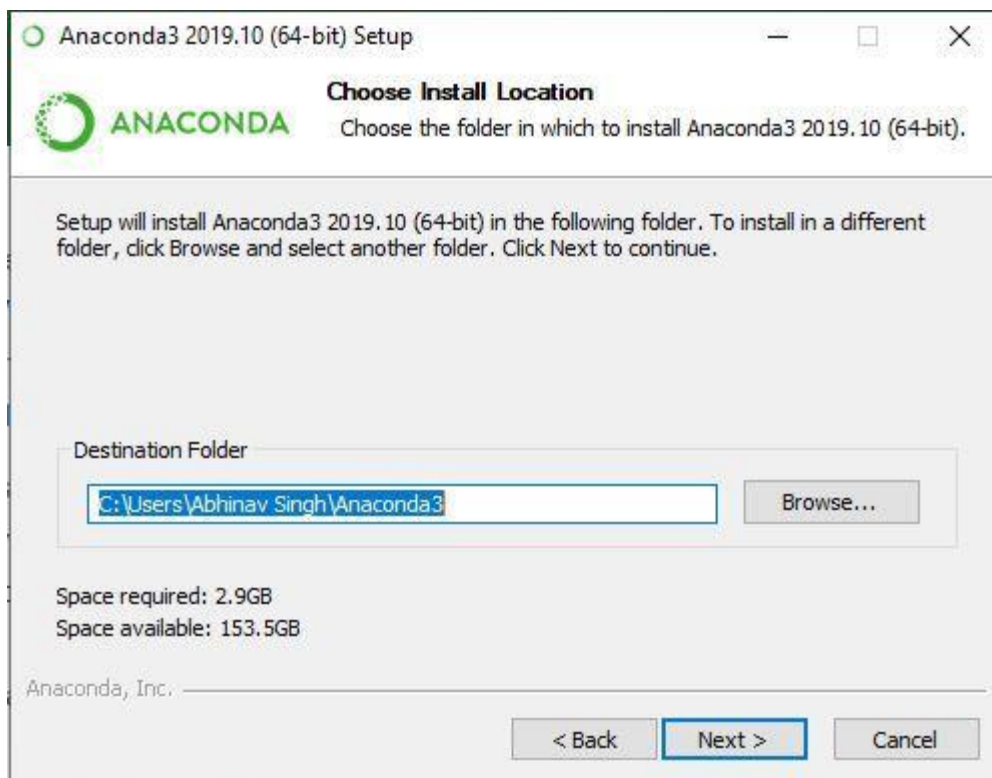
- **Getting through the License Agreement:**



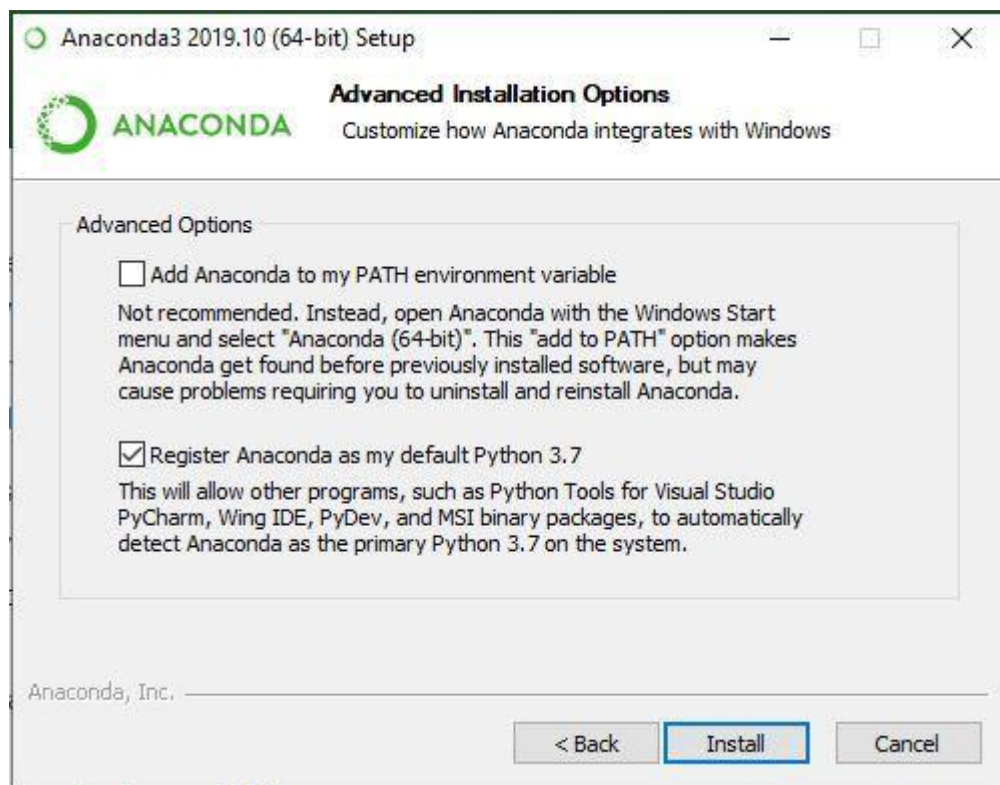
- Select Installation Type: Select Just Me if you want the software to be used by a single User



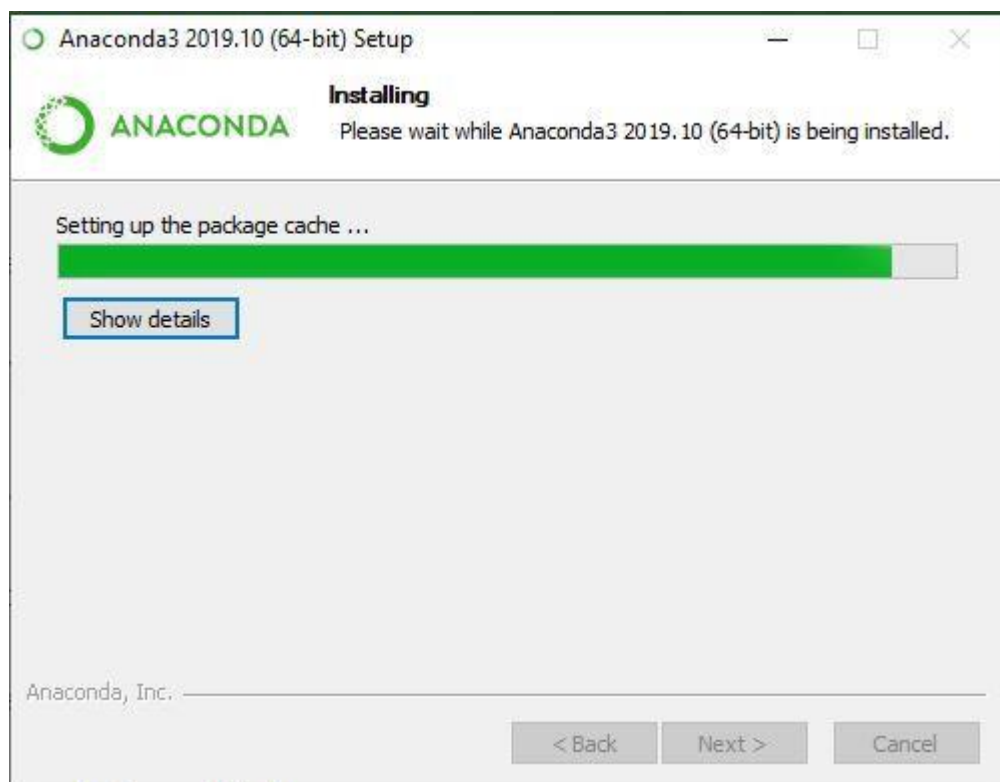
- Choose Installation Location:



- Advanced Installation Option:



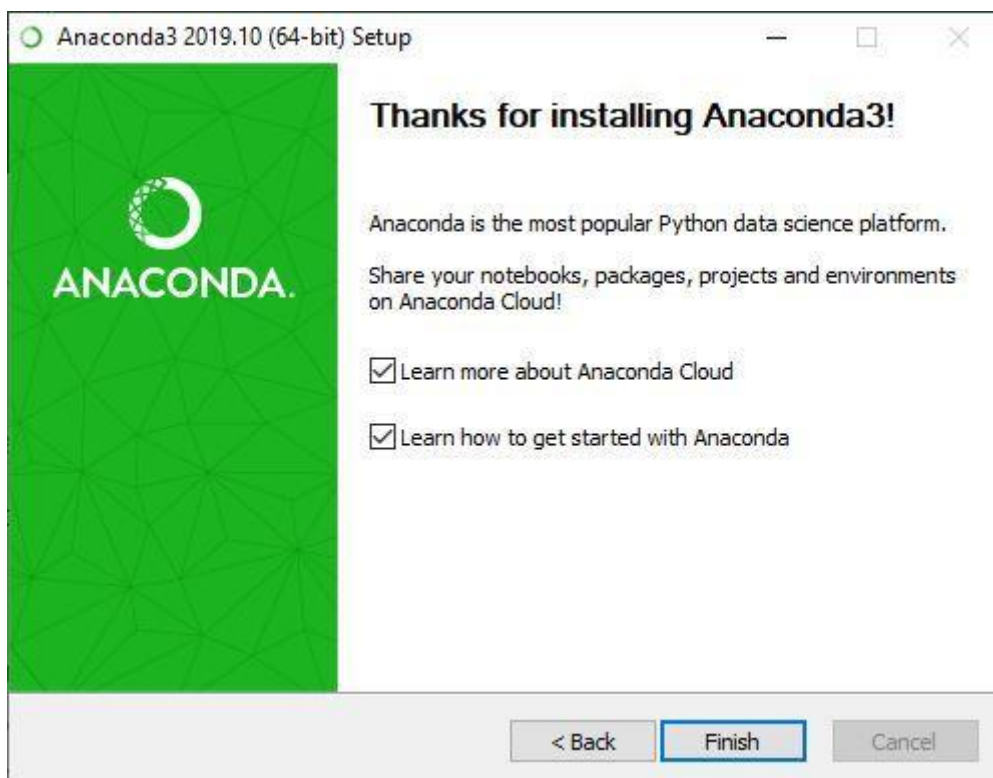
- **Getting through the Installation Process:**



- **Recommendation to Install Pycharm:**



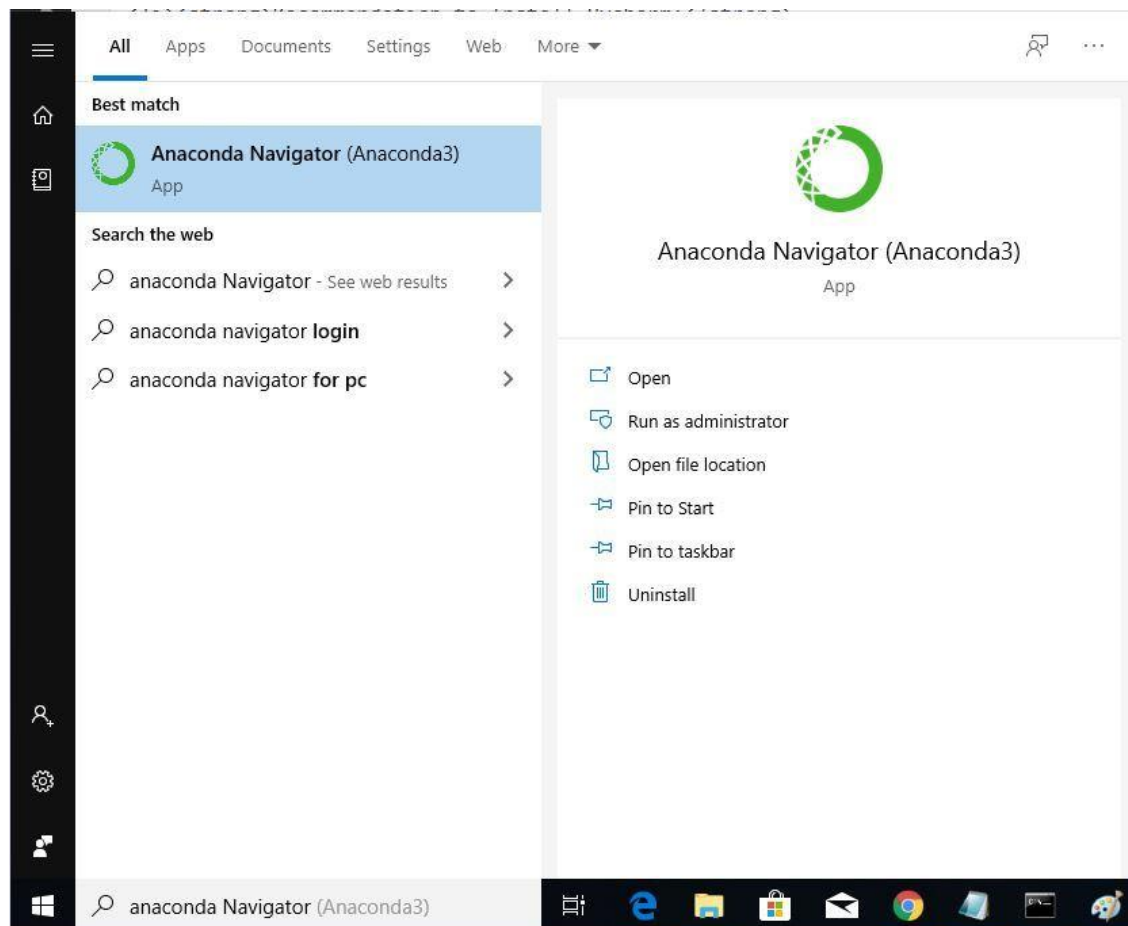
- **Finishing up the Installation:**



- **Working with Anaconda:**

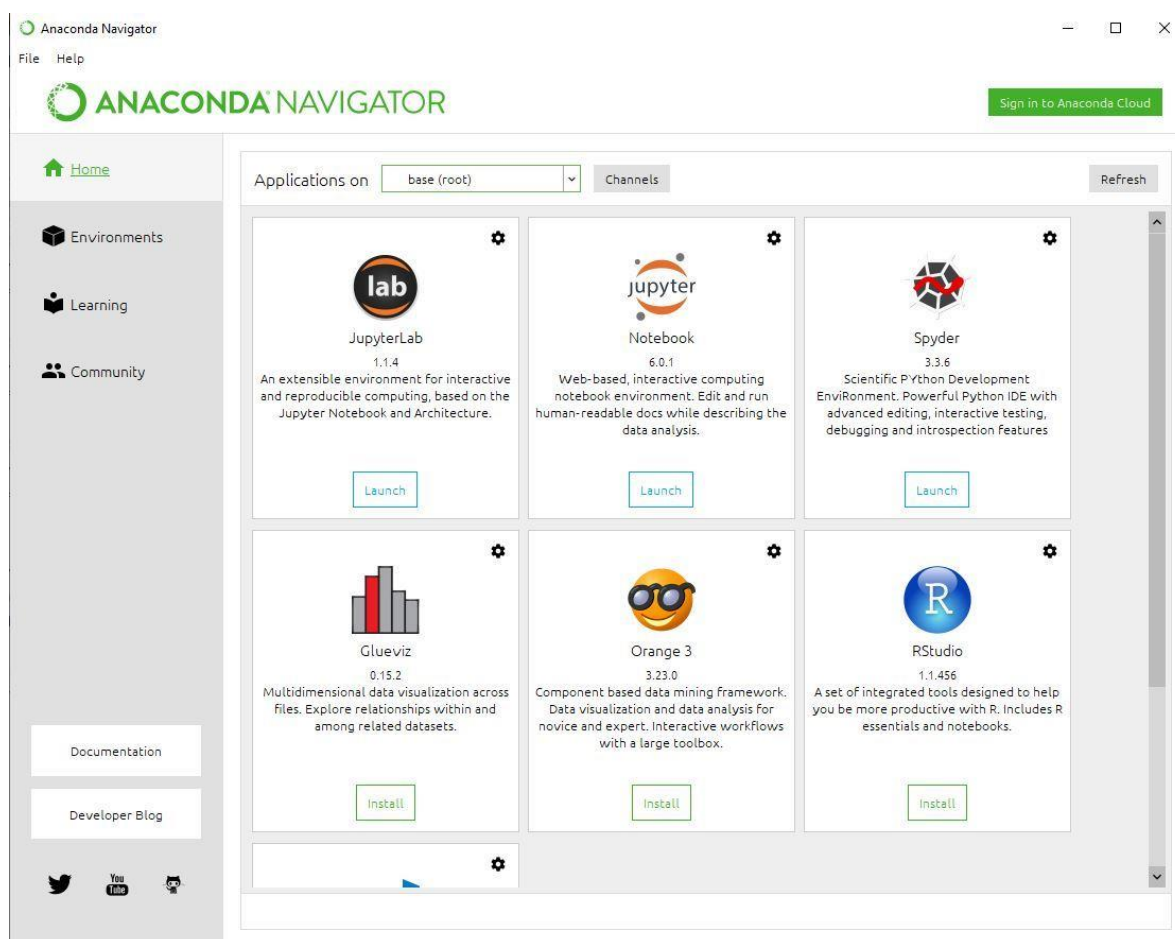
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- Once the installation process is done, Anaconda can be used to perform multiple operations. To begin using Anaconda, search for Anaconda Navigator from the Start Menu in Windows





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Tool based Lab Manual

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Preview

Python is an elegant and robust programming language that delivers both the power and general applicability of traditional compiled languages with the ease of use (and then some) of simpler scripting and interpreted languages. Python is a general-purpose interpreted, interactive, object-oriented, and high-level programming language.

Work on Python began in late 1989 by Guido van Rossum, then at CWI (Centrum voor Wiskunde en Informatica, the National Research Institute for Mathematics and Computer Science) in the Netherlands.

Python source code is also available under the GNU General Public License (GPL).

Python is processed at runtime by the interpreter. You do not need to compile your program before executing it.

Python is a great language for the beginner-level programmers and supports the development of a wide range of applications from simple text processing to WWW browsers to games



Python Installation

1. Download the current production version of Python (3.10) from the <https://www.python.org/>
2. Double click on the icon of the file that you just downloaded.
3. Select Add path automatically option
4. Proceed until you get to the Finish button. Your installation is complete.

Setting up the Environment

1. Starting at My Computer go to the following directory C:\Python37. In that folder you should see all the Python files.
2. Copy that address starting with C: and ending with 27 and close that window.
3. Click on Start. Right Click on My Computer.
4. Click on Properties. Click on Advanced System Settings or Advanced.
5. Click on Environment Variables.
6. Under System Variables search for the variable Path.
7. Select Path by clicking on it. Click on Edit.
8. Scroll all the way to the right of the field called Variable value using the right arrow.
9. Add a semi-colon (;) to the end and paste the path (to the Python folder) that you previously copied. Click OK.

Open command prompt and type “python” (not in double quote)

Creating a python programming file

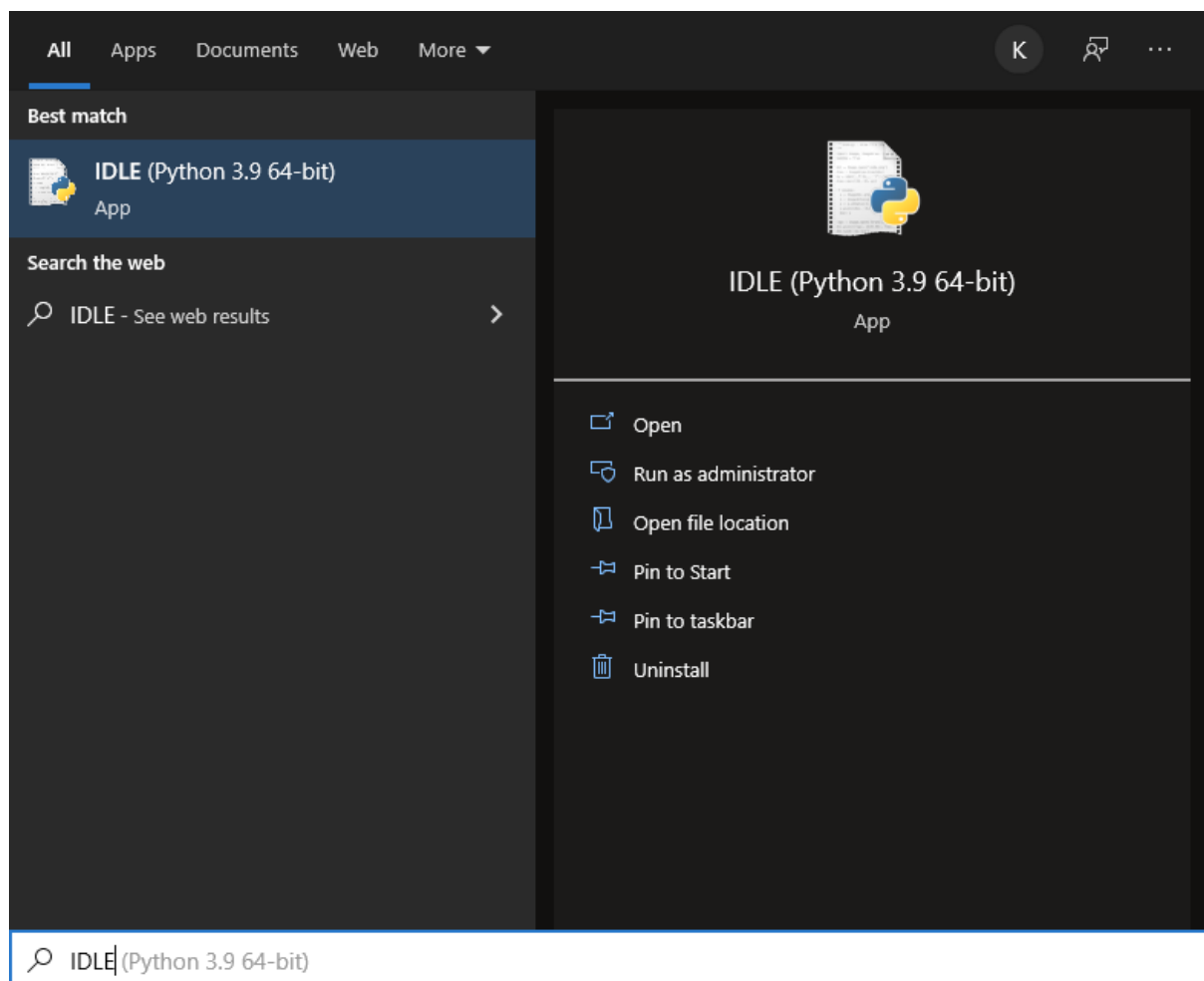
Tools required: Python(3.10)

Table to Download links for tools:

Python	https://www.python.org/downloads/
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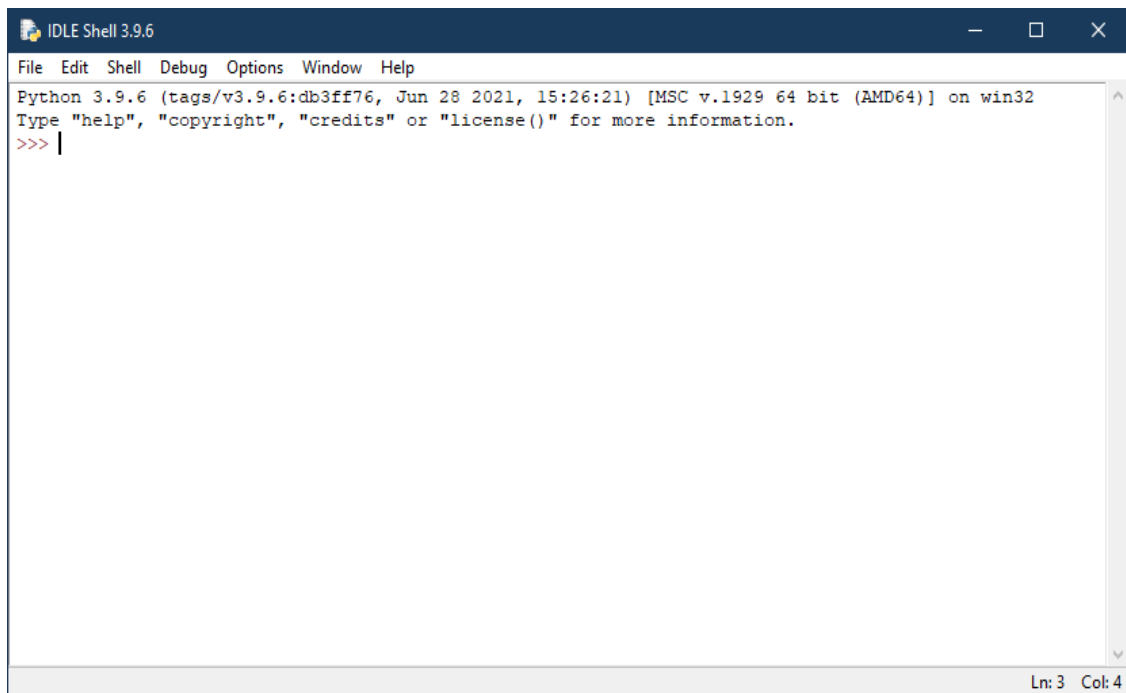
Creating our first R program:

1. Search IDLE in start menu



2. Open IDLE python, your console should look like given below figure.

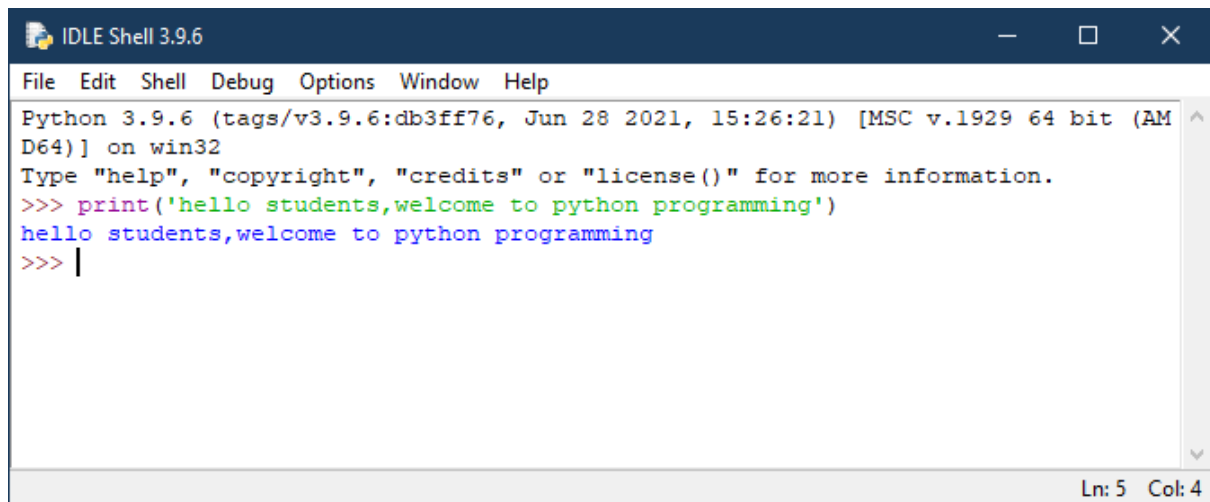
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```
IDLE Shell 3.9.6
File Edit Shell Debug Options Window Help
Python 3.9.6 (tags/v3.9.6:db3ff76, Jun 28 2021, 15:26:21) [MSC v.1929 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> |
```

3. Now let us print string.

- Type `print('hello students ,welcome to python programming')`
- Press enter.



```
IDLE Shell 3.9.6
File Edit Shell Debug Options Window Help
Python 3.9.6 (tags/v3.9.6:db3ff76, Jun 28 2021, 15:26:21) [MSC v.1929 64 bit (AM
D64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> print('hello students,welcome to python programming')
hello students,welcome to python programming
>>> |
```

(Figure shows print function)

- Save the file, click on file→Save→select the destination→write name of file→click save. (Or press CTRL+S)

➤ IDE

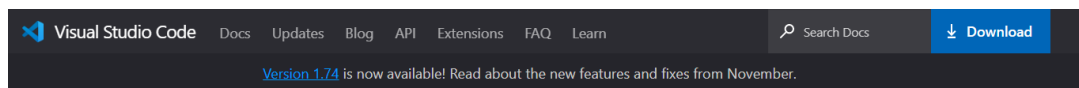
- IDE stands for **Integrated development environment**.
- It is nothing more than an enhanced version of a text editor that helps you write more efficient and nicer code.
- It helps to differentiate different parts of your codes with different colors and notifying you if you are missing some semicolon or bracket at some place by highlighting that area.
- A lot of IDEs are available such as **DEV++ Code Blocks, VS Code**.

➤ Visual Studio Code

- Getting up and running with Visual Studio Code is swift and straightforward.
- VS Code is a free text editor created by Microsoft.
- It is a good editor for JavaScript, HTML and CSS.
- It is a small download so you can install it quickly.
- Additionally, it runs on the Windows, macOS, and Linux operating systems.
- It also supports a large number of extensions that allow the VS Code editor to be adapted to many purposes.

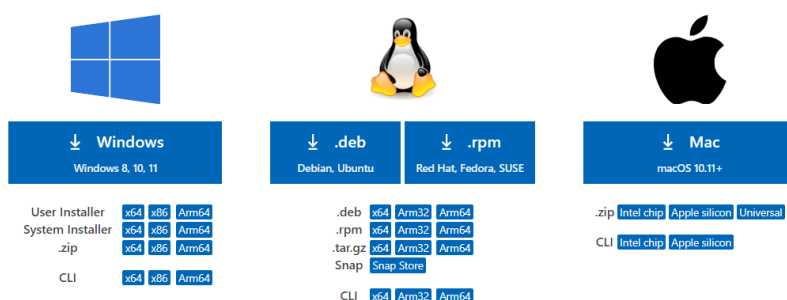
➤ Visual Studio Code Installation

- Visit <https://code.visualstudio.com/download>



Download Visual Studio Code

Free and built on open source. Integrated Git, debugging and extensions.



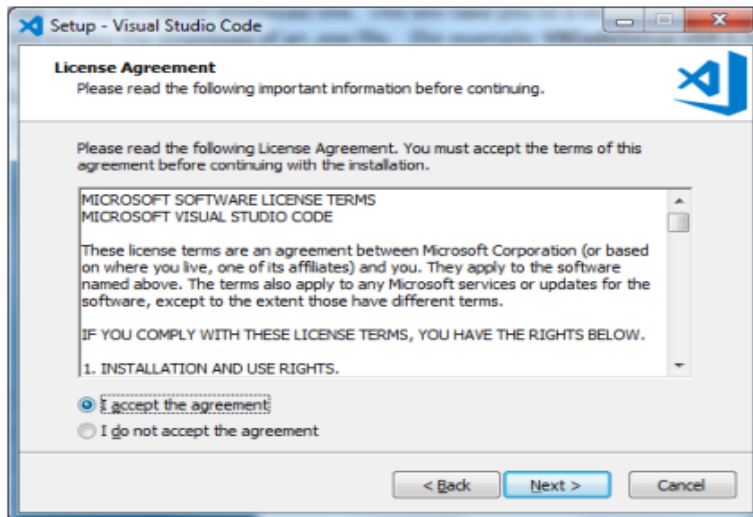
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- Click on the Windows download link. This will take you to a documentation page and also initiate the download of an .exe file.
- Run the exe. Step through the installation wizard as indicated below.

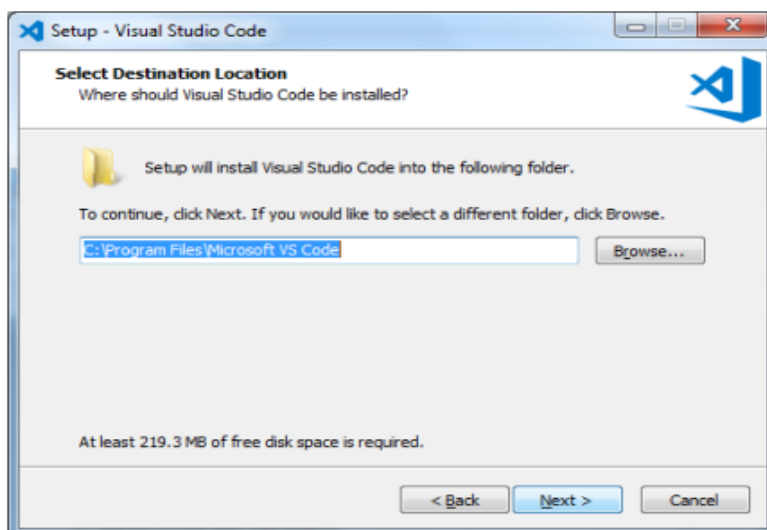


Click Next

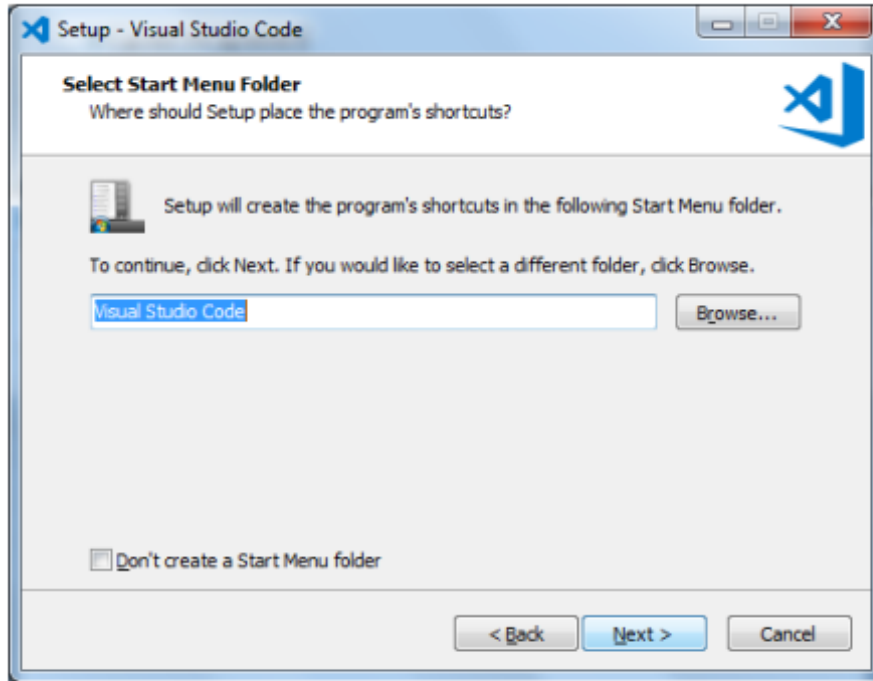
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Accept the license agreement.

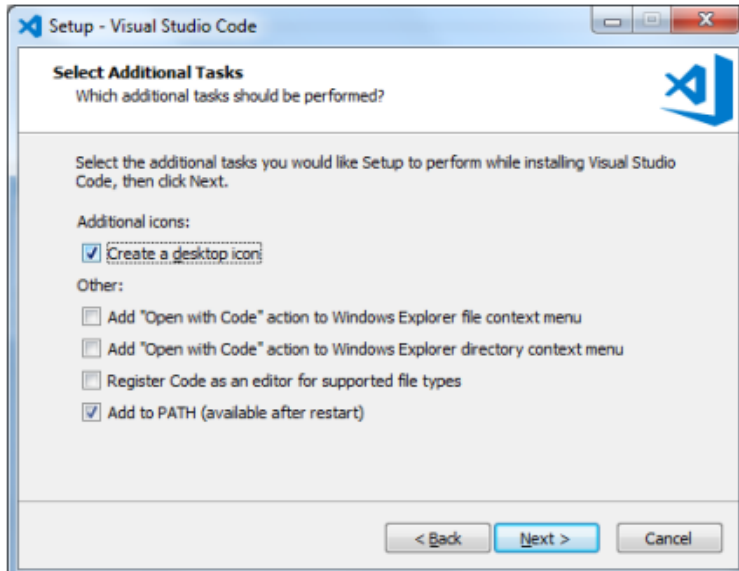


Use the default installation folder. Click Next.

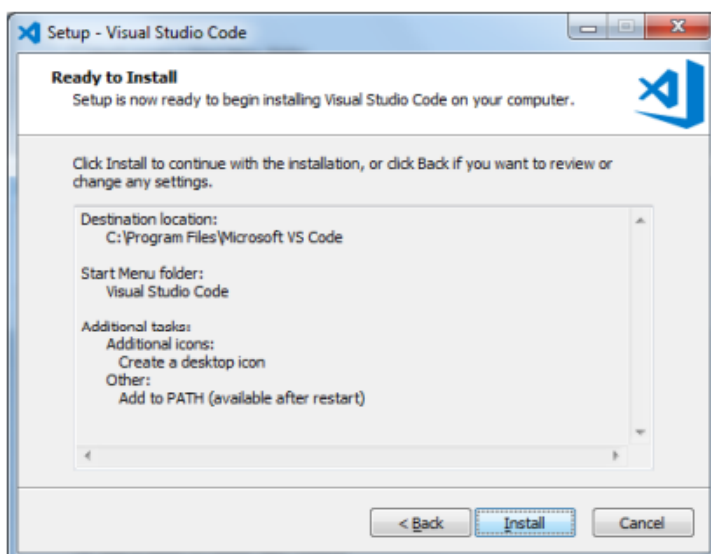


Click Next

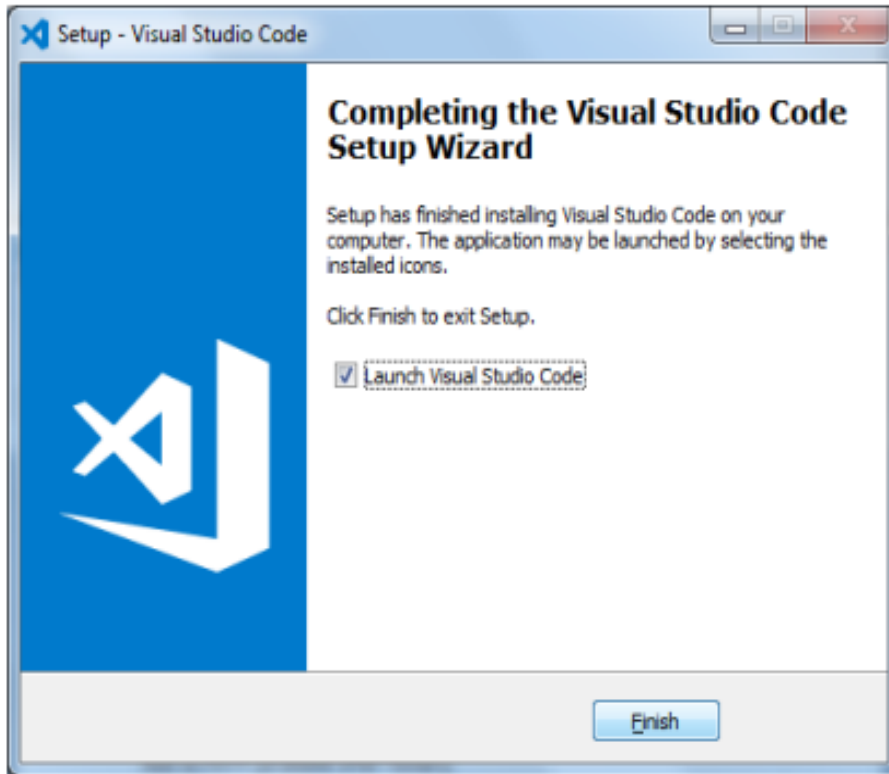
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Creating a desktop shortcut is recommended. You will use VS Code a lot in this course, and you want it to be easy to find. Check the boxes as shown. Click Next

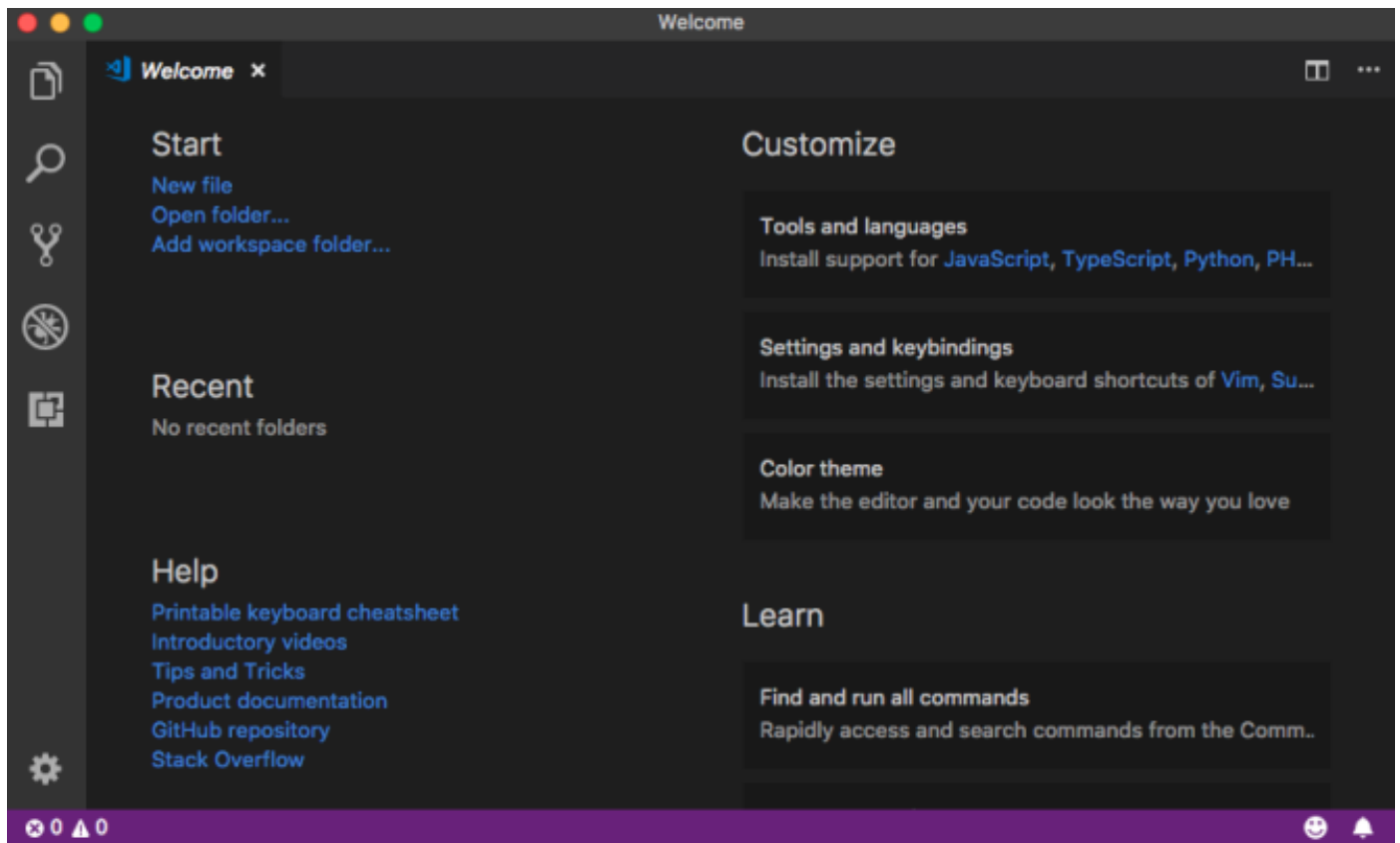


Click Install.



Click Finish.

- VS Code will start, and you will see a screen similar to the following:



- On the left side, you should see a couple of icons. One of them (under the no bugs icon) is the extensions button:

➤ Extensions:

Keyboard Shortcut: `Ctrl+Shift+X`

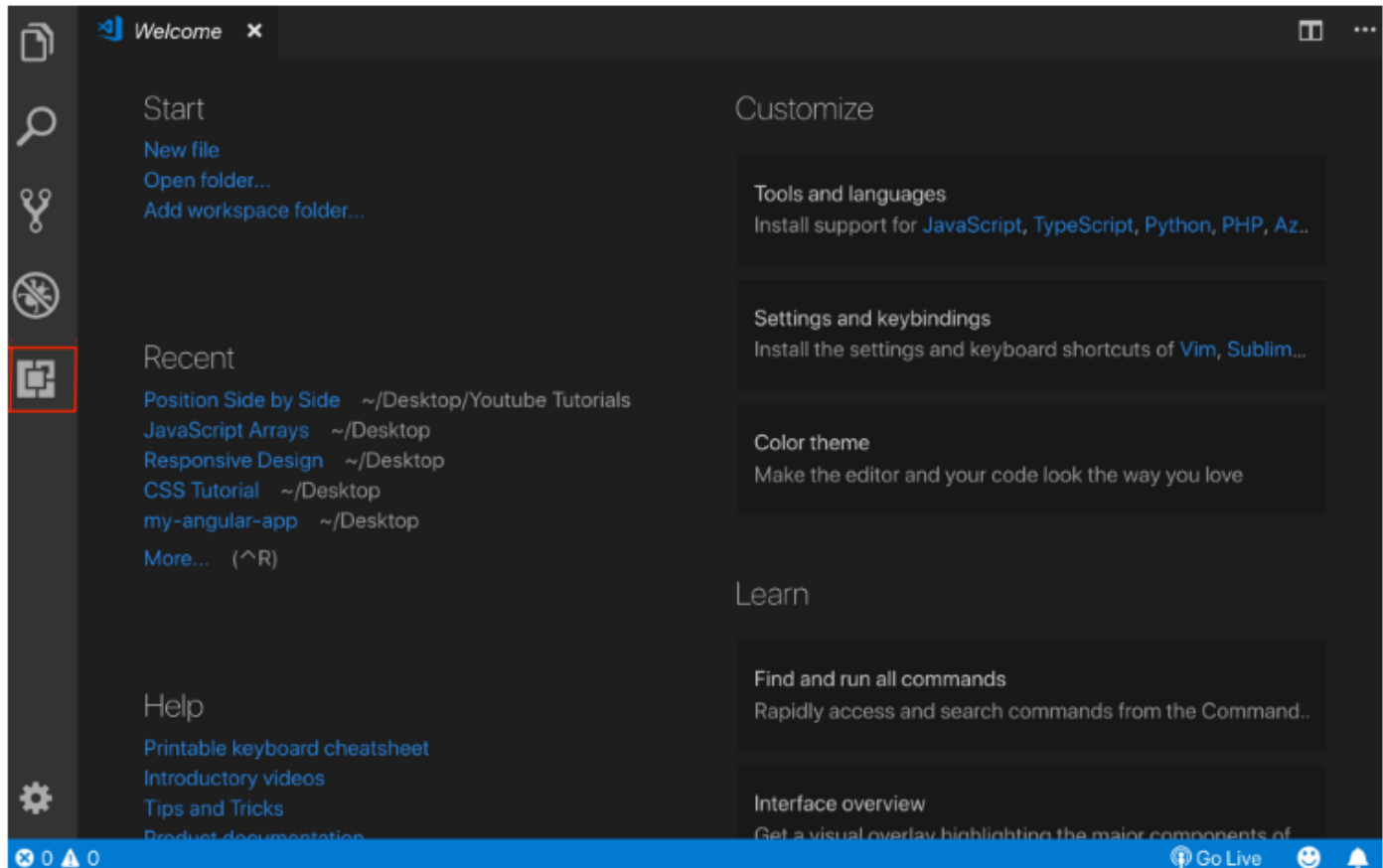
Find extensions:

1. In the VS Code Marketplace.
2. Search inside VS Code in the **Extensions** view.
3. View extension recommendations
4. Community curated extension lists, such as awesome-vscode

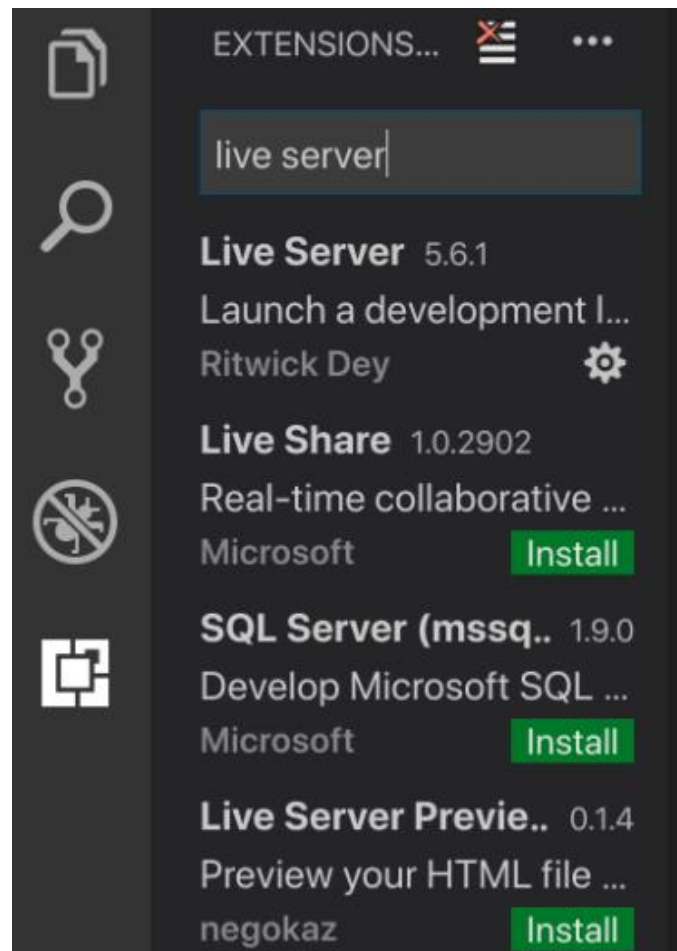
Enable additional languages, themes, debuggers, commands, and more. VS Code's growing community shares their secret sauce to improve workflow.

Installing Extensions:

In the **Extensions** view, you can search via the search bar or click the **More Actions (...)** button to filter and sort by install count.



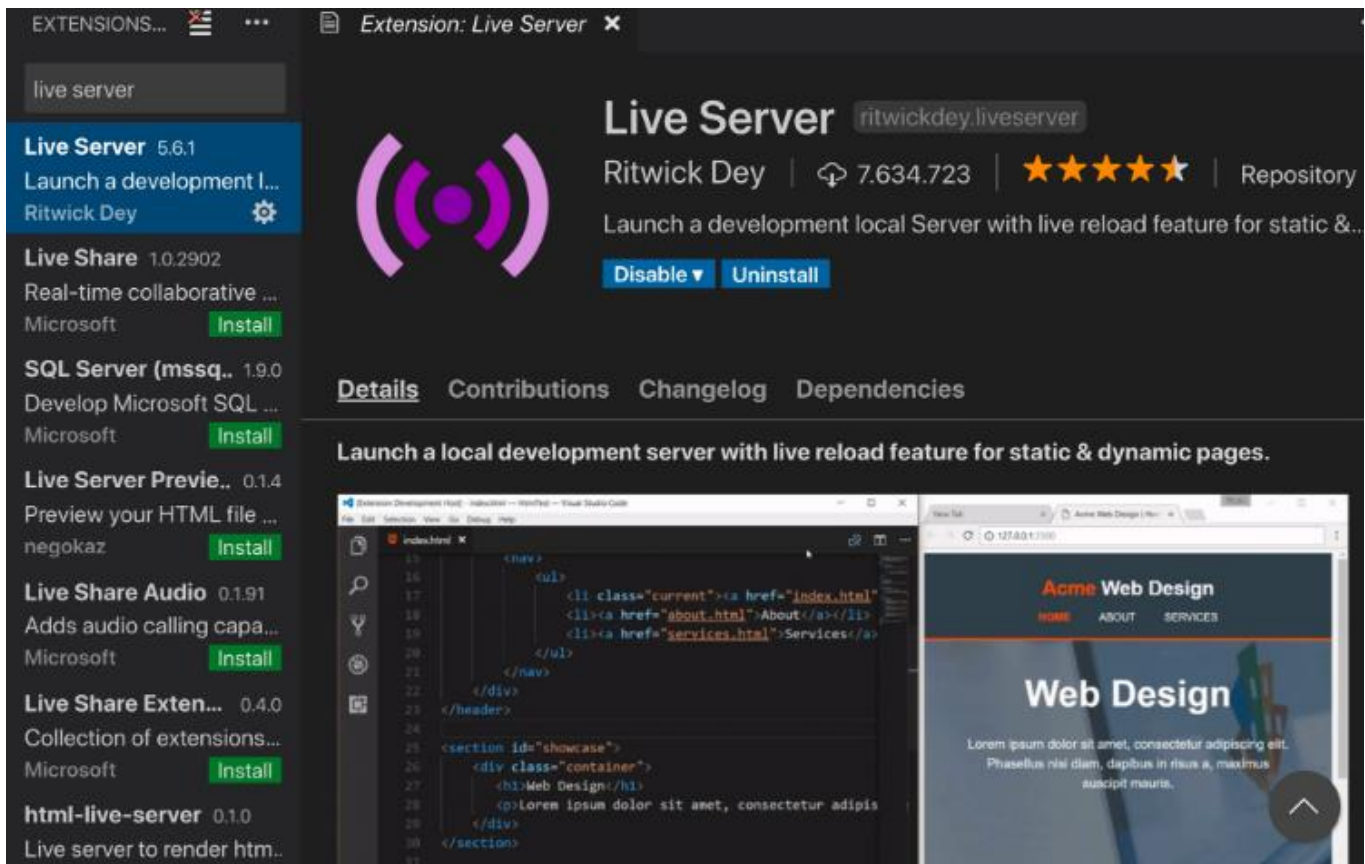
- Once you click on it a search bar will appear. Just type in **“live server”**.



➤ Live Server installation

- You will see many options, so you can choose whichever one works for your system.

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The screenshot displays the Visual Studio Code interface with the 'Live Server' extension installed. The extension is by Ritwick Dey and has a version of 5.6.1. It is described as a tool to 'Launch a development local Server with live reload feature for static &..'. The interface also shows a list of other extensions on the left, including 'Live Share', 'SQL Server (mssql..)', 'Live Server Preview..', 'Live Share Audio', 'Live Share Exten...', and 'html-live-server'.

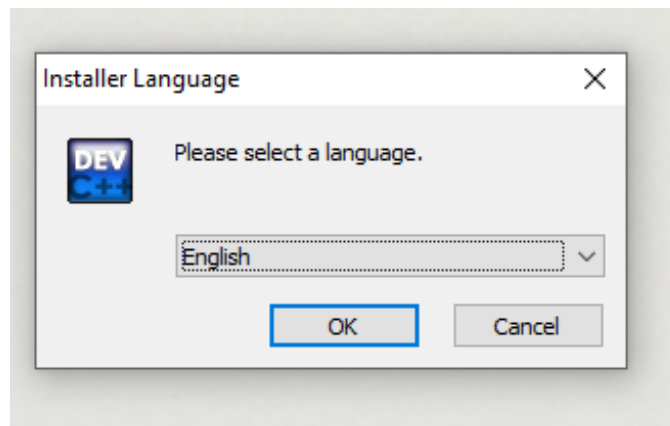
➤ Why Live Server

- Normally, when you make a change in your code or write something new, you need to refresh the page manually to see the changes.
- In other words, if you make 100 changes in your code each day, you need to refresh the browser 100 times.
- The live-server extension, however, automates this for you. After installing it, an automated localhost will be able to run in your browser, which you can start with a single button.
- Once you make changes in your code or write something new, after saving it, the browser will auto-refresh itself. Then you will be able to see the changes quickly and automatically.

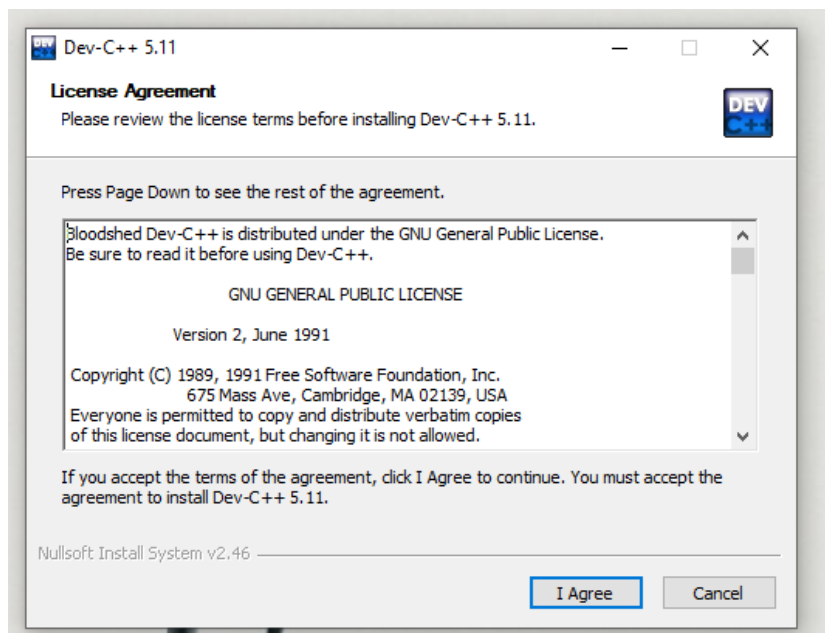
Installation and Getting Started under Windows OS

Follow the steps mentioned in following figure to install Dev c++ on Windows Operating System.

1. Go to <https://sourceforge.net/projects/orwelldvcpp/>
2. Click on Download and wait for five seconds.
3. Double click on Dev-Cpp 5.11 TDM-GCC 4.9.2 Setup file

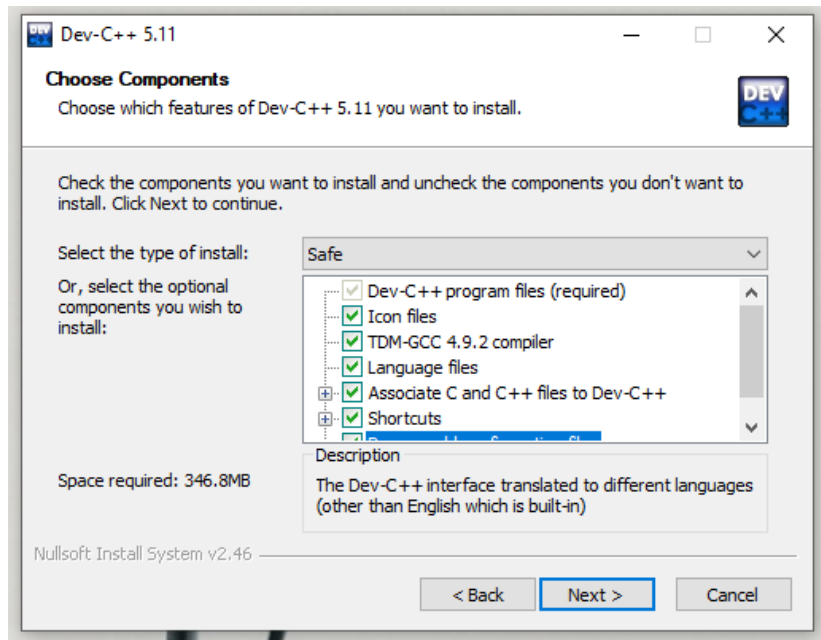


4. Click On Ok.

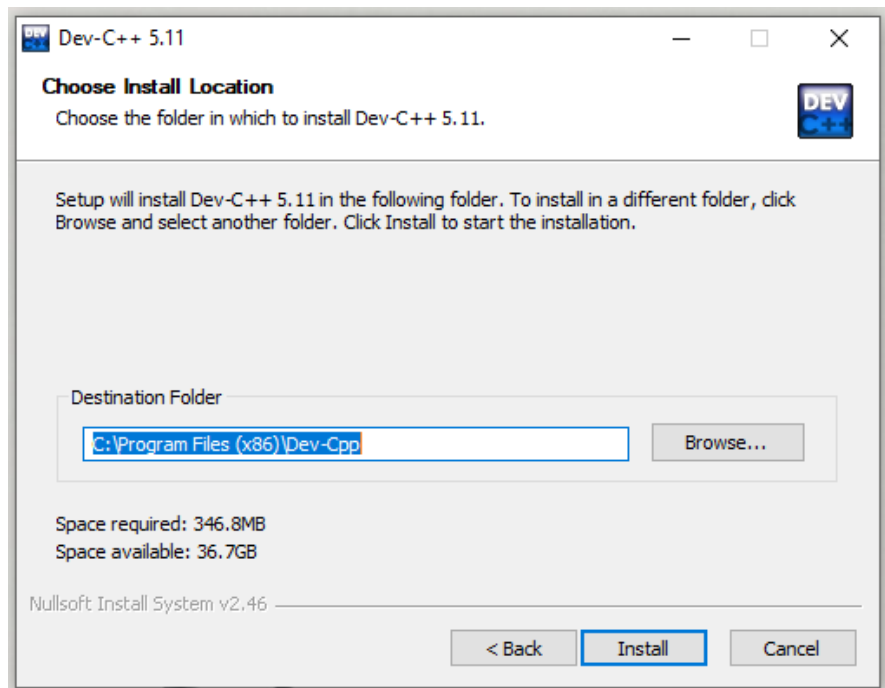


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5. Click On I Agree.

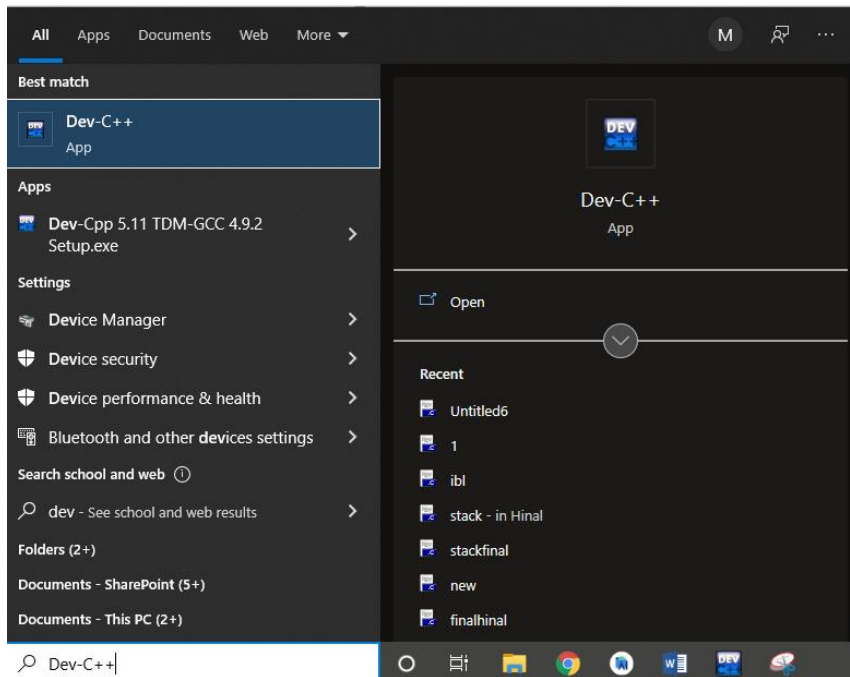


6. Click on Next.

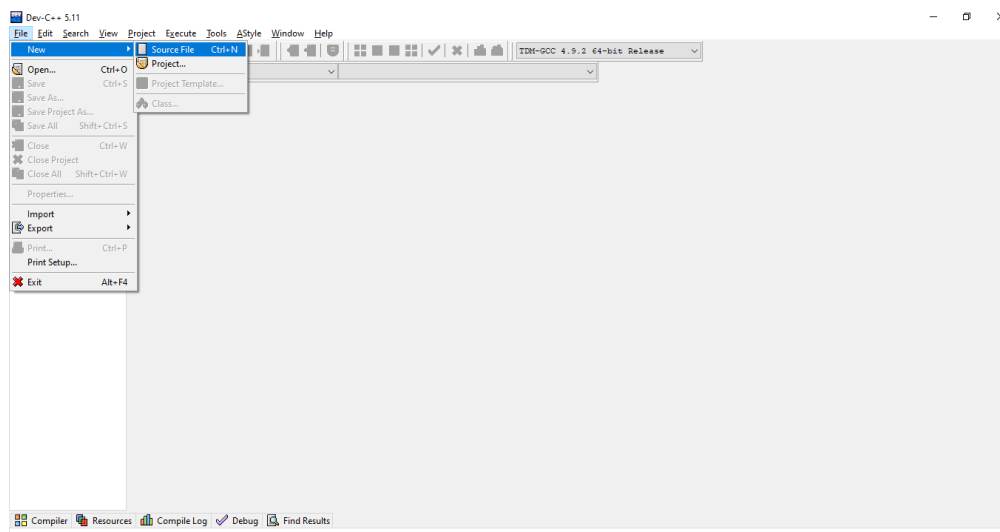


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9. Click on Next, Next Okay.

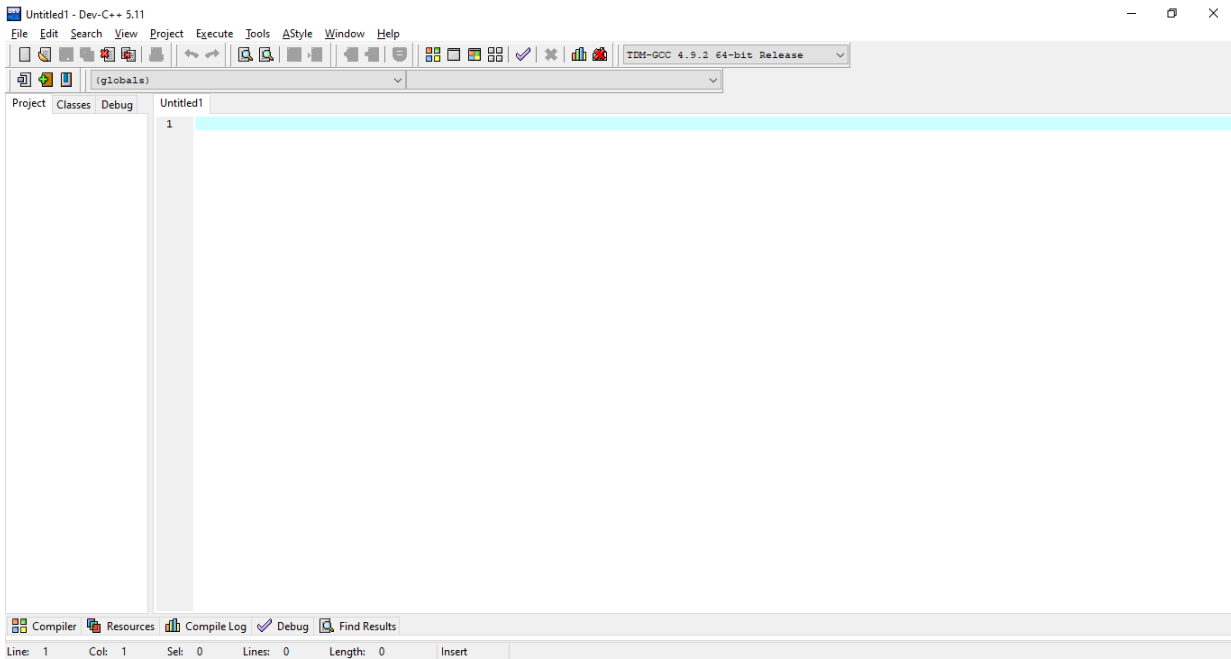


10. Open Dev C++.

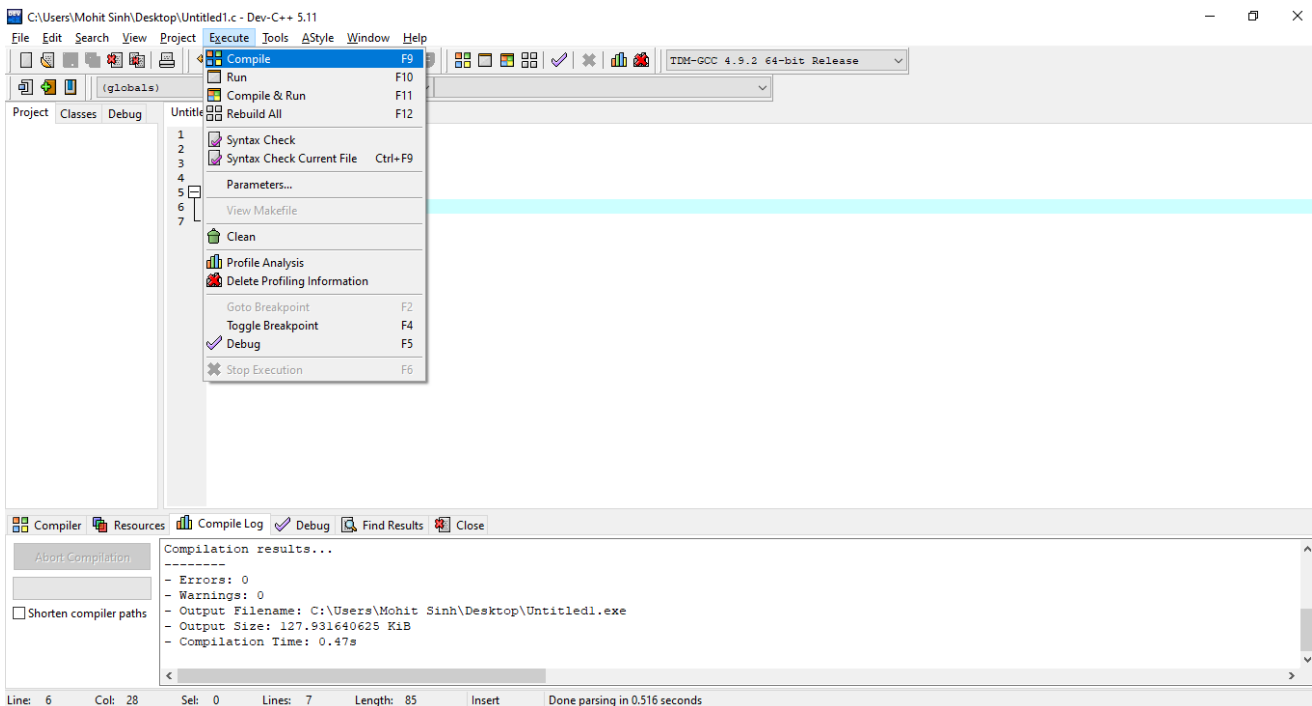


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11. IDE of Dev C++.

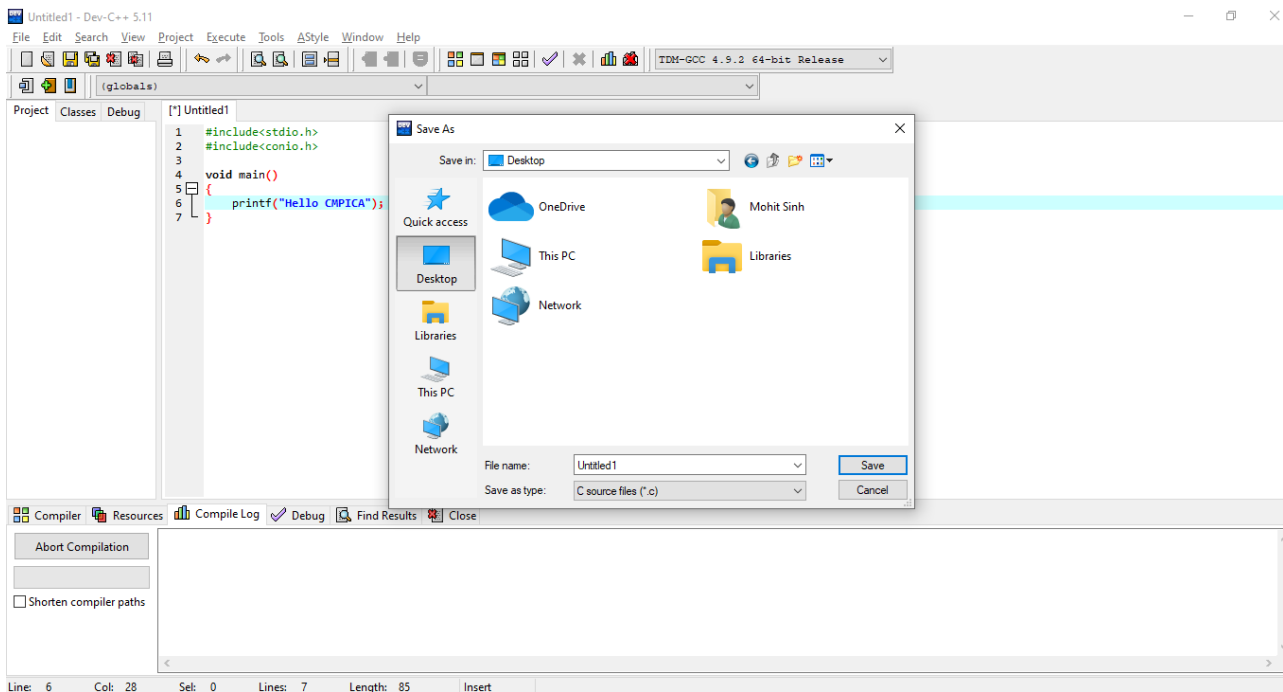


12. New file for c programming.

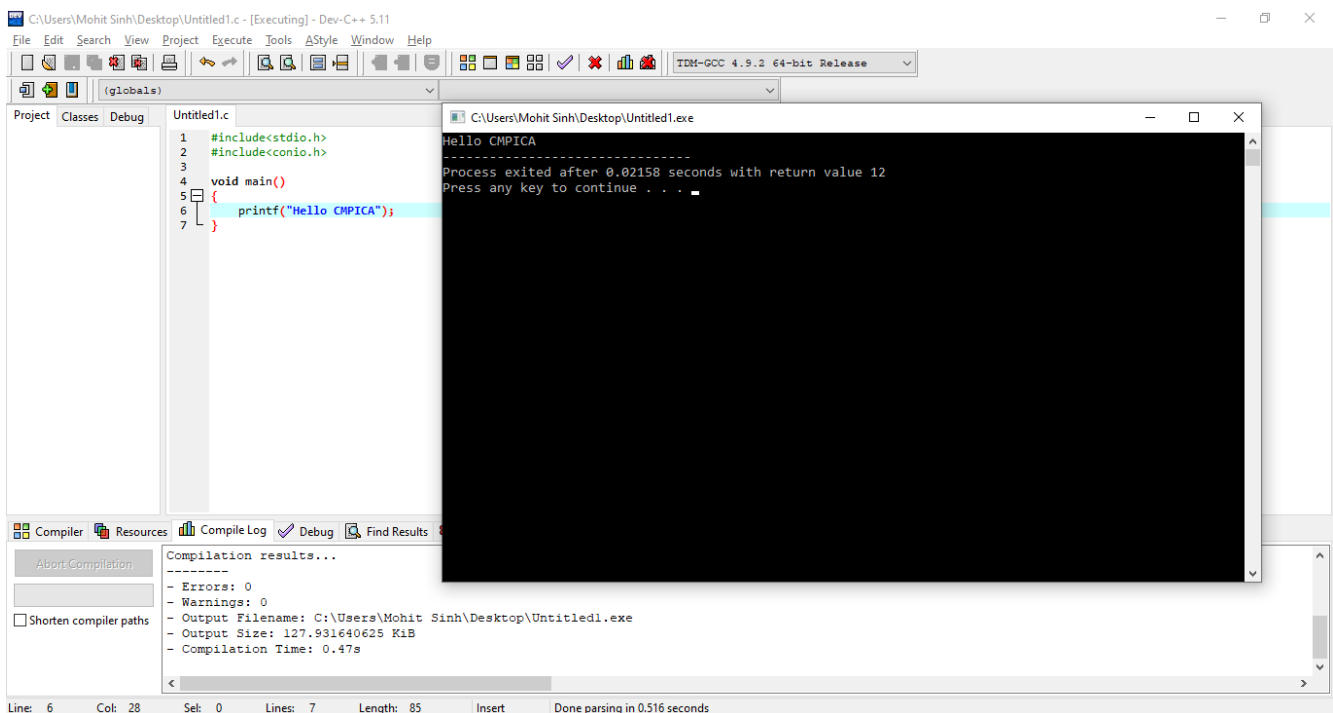


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13. From the execute tab of menu bar you have to select compile option.



14. To run the program, save the file with appropriate name and extension.



The black screen window is the console window that display the output of the program.

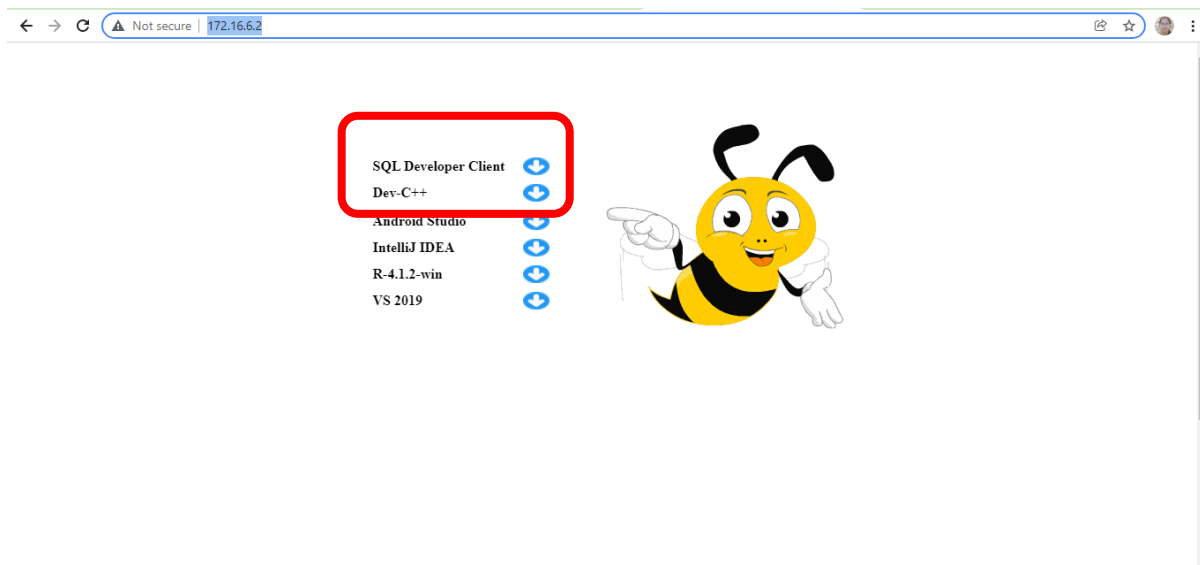
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Objectives

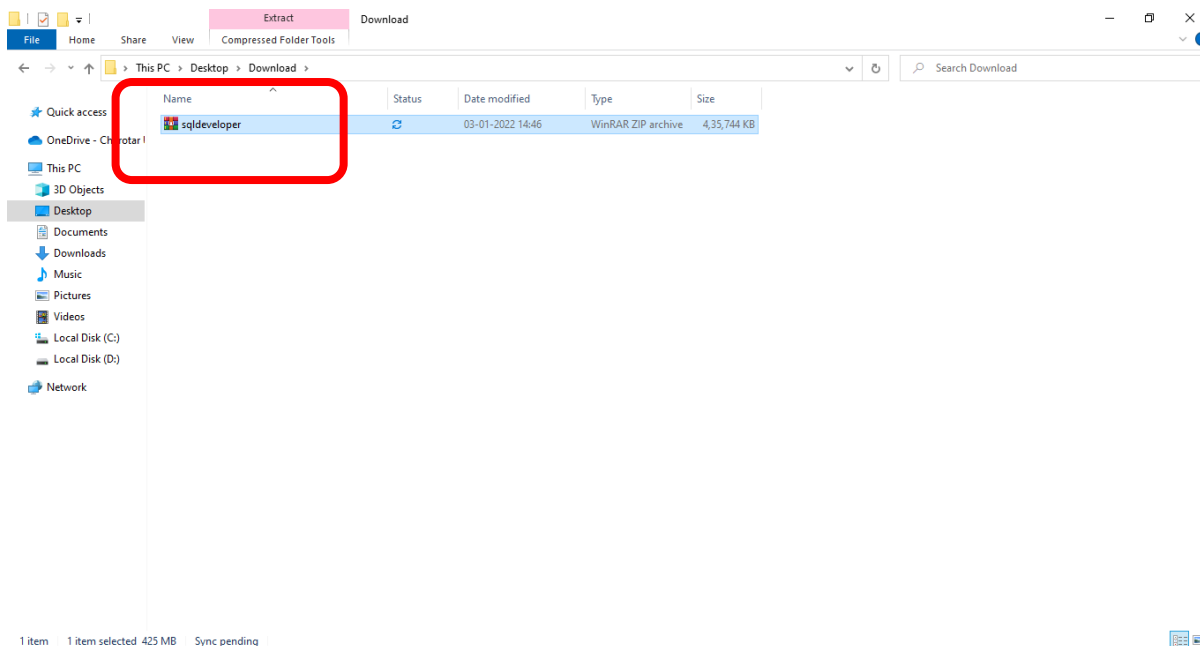
- To make students acquaint with installation of Database Client Software.

Step: 1 Go to <http://172.16.6.2/>

Download SQL Developer Client

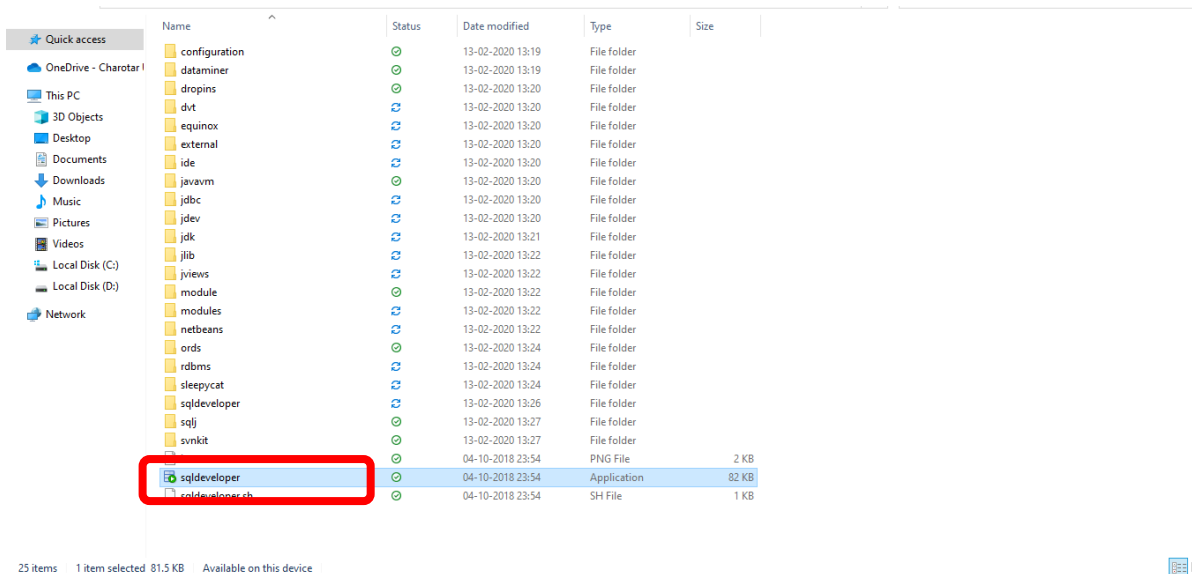


This will download the zip with name **sqldeveloper**. Unzip it.

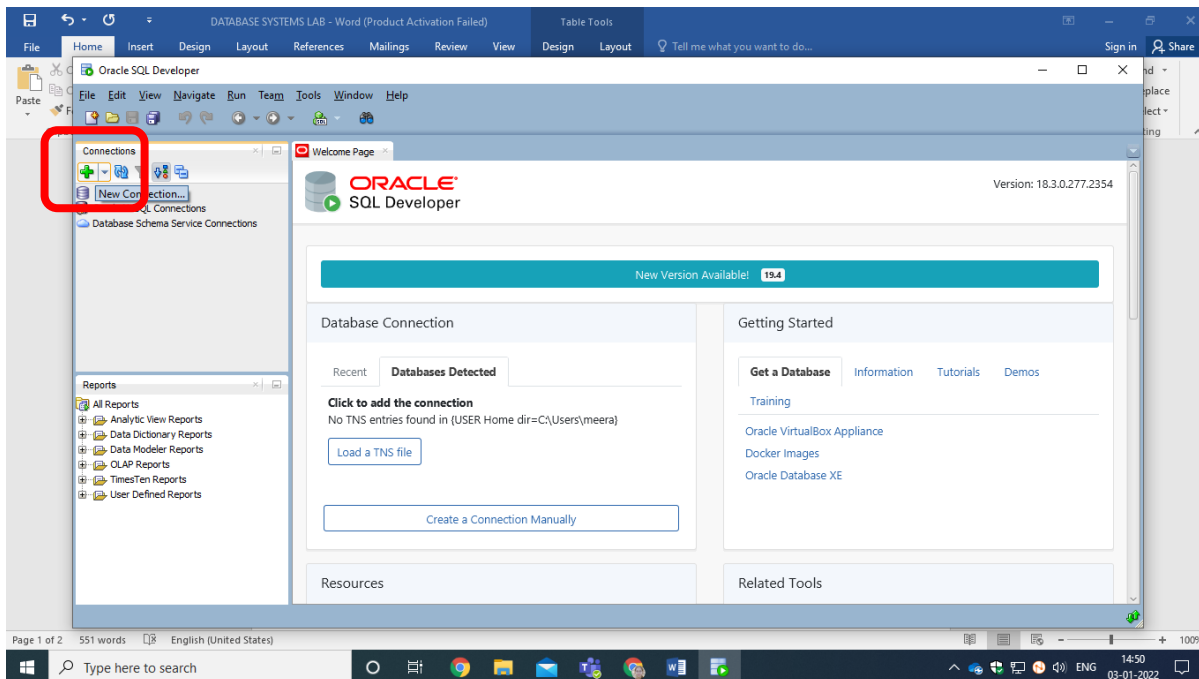


Step: 2 Double click on **sqldeveloper** Application

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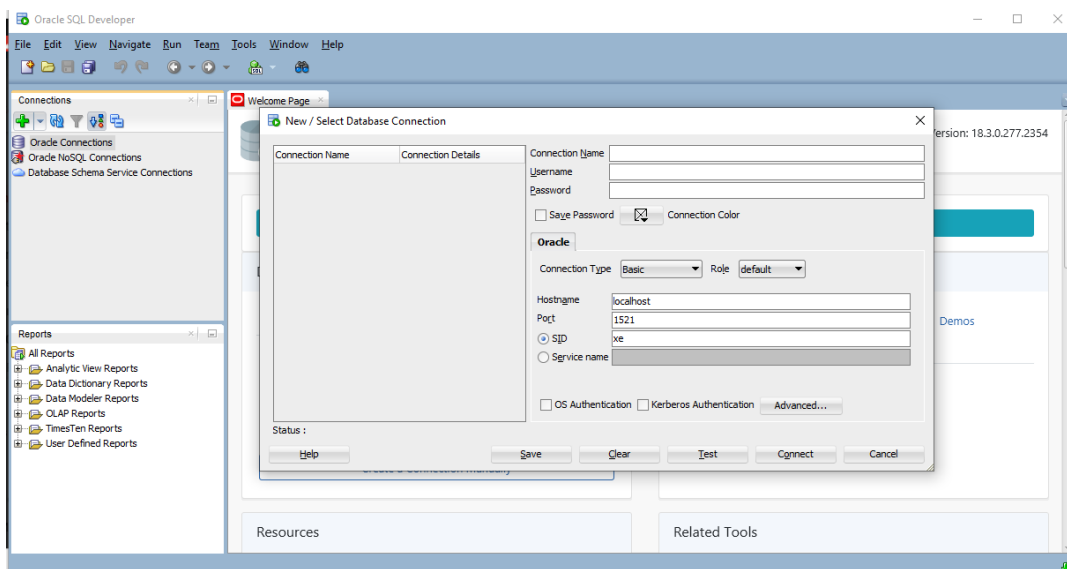


It will open the Oracle SQL Developer.



Step: 3 Click on  Sign, the following screen will appear.

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Step: 4 Give Proper Connection Name for example CMPICA

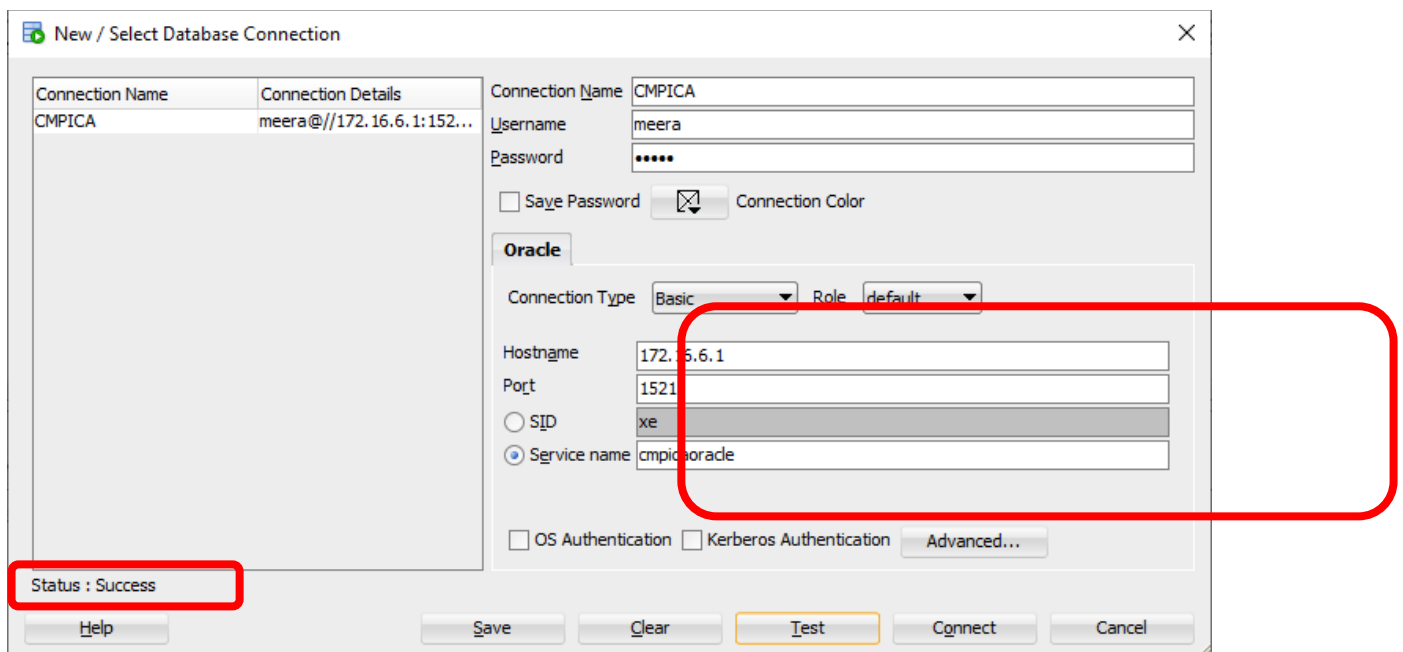
Enter Username and Password provided to you in respective fields.

In Hostname write **172.16.6.1**

Click service name radio button and enter **cmpicaoracle**

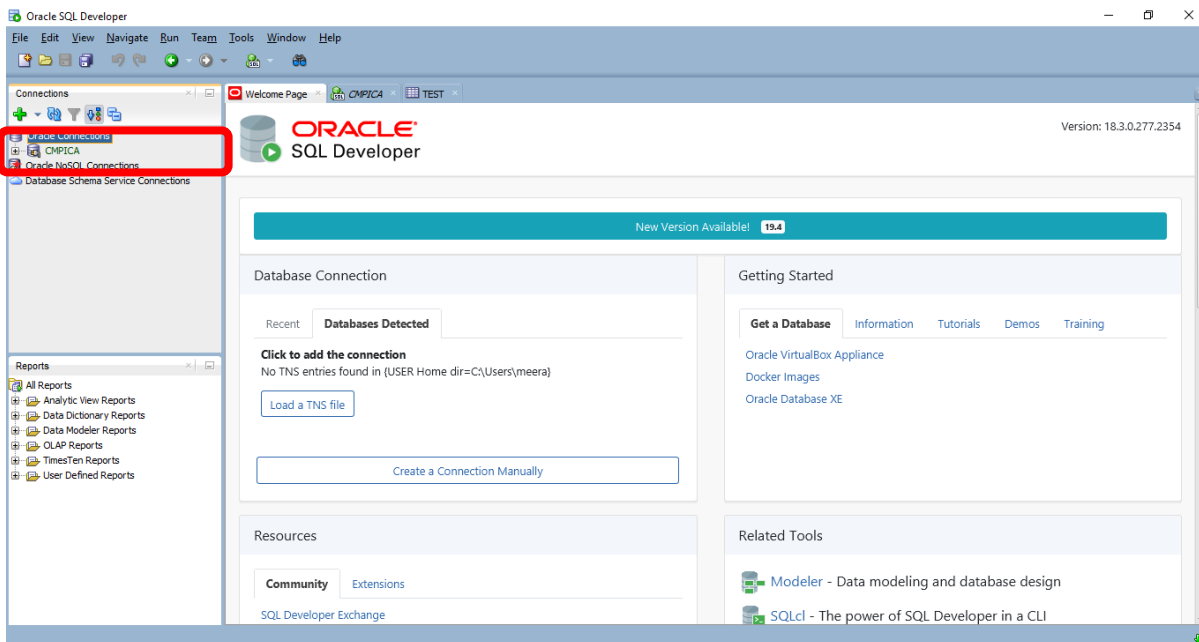
KEEP REST OF THE CONFIGURATION AS IT IS. And Click on Test button. The **success** should get displayed.

Now click on Connect Button.

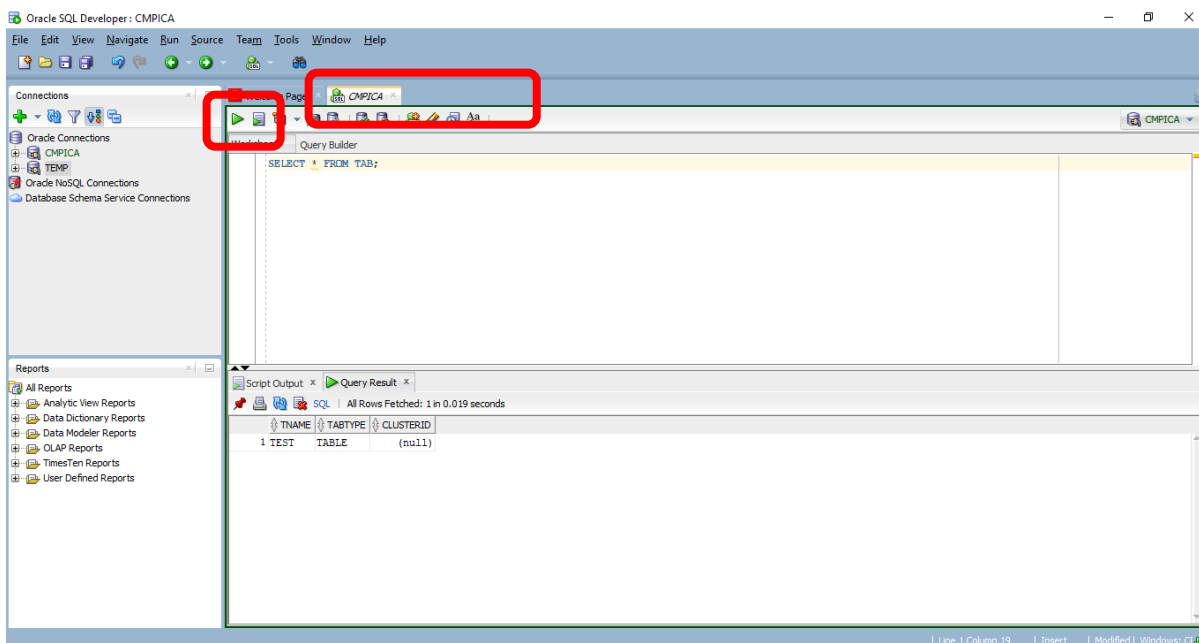


Your newly created connection should get displayed on the left panel.

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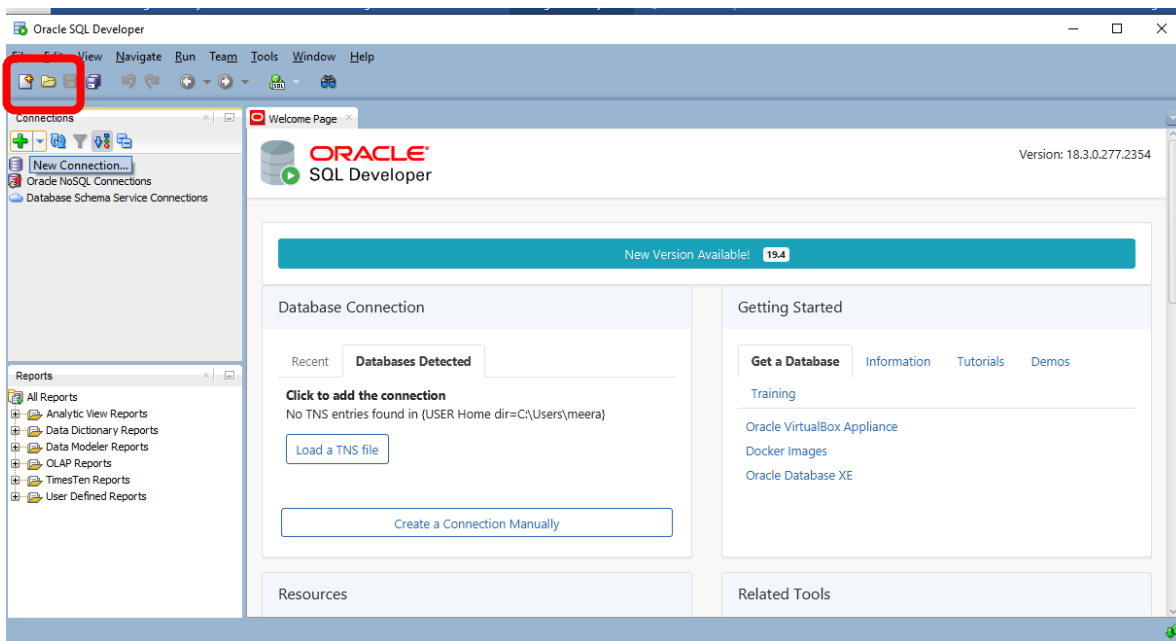
Step 5 On the right side panel observe the Tab with CMPICA (Connection Name). Write your query and Run your query (execute) using  icon **or** press Ctrl + Enter.



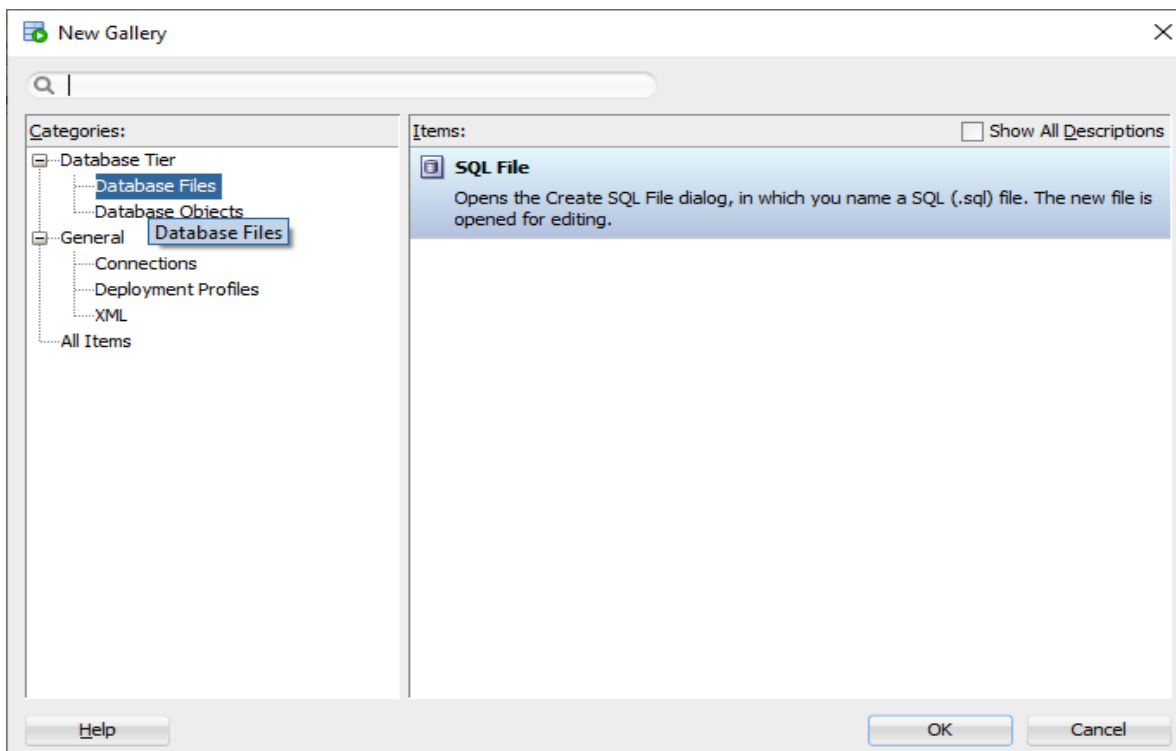
Step 6 IF you want to create a new SQL file then

In the right panel click on File -> New symbol,  or press Ctrl + N and Then select database -> SQL File and click OK. Please refer below screenshots.

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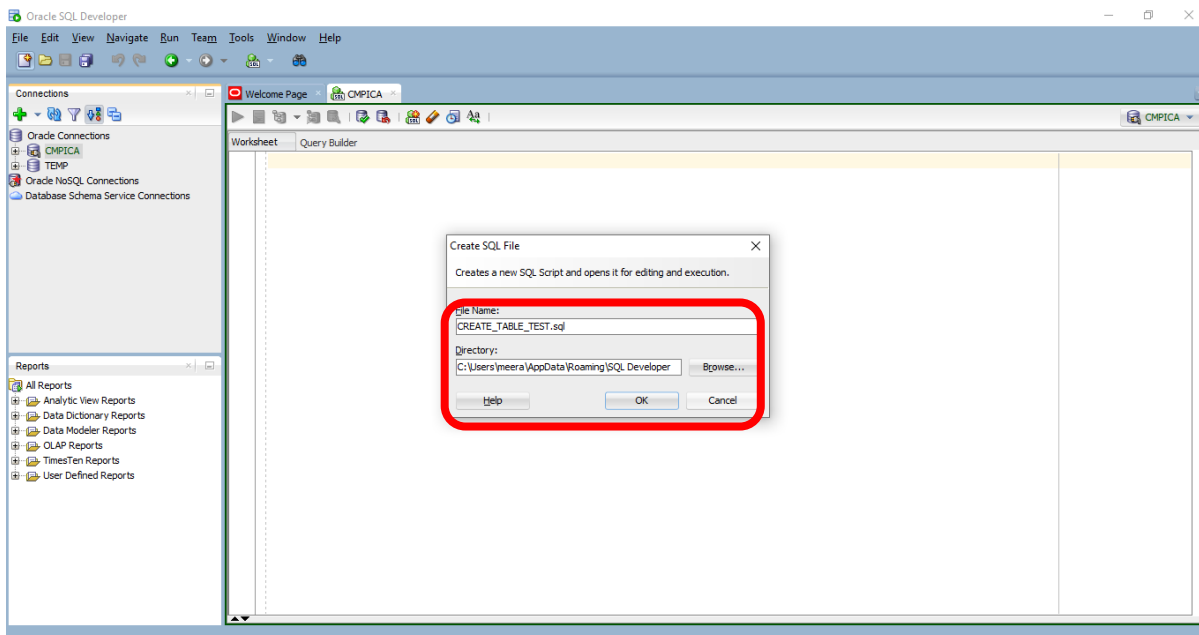


The following screen will appear.



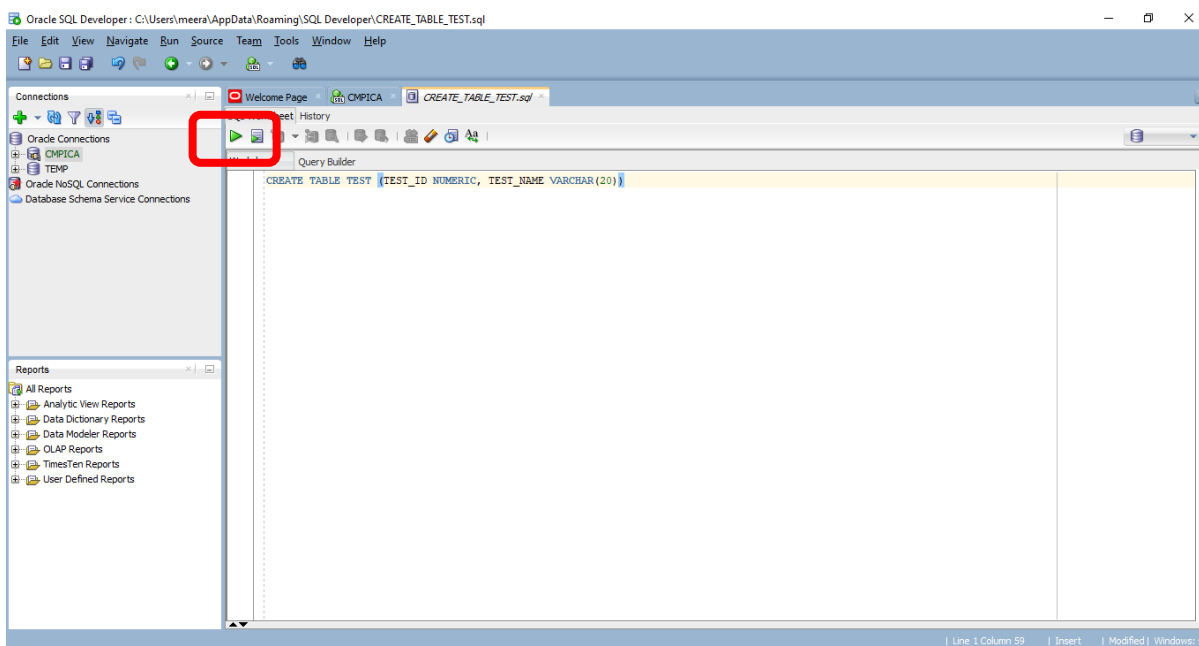
Give proper file name and save to the proper location. **[IT IS ADVISED TO SAVE ALL THE SQL IN ONE FOLDER]**

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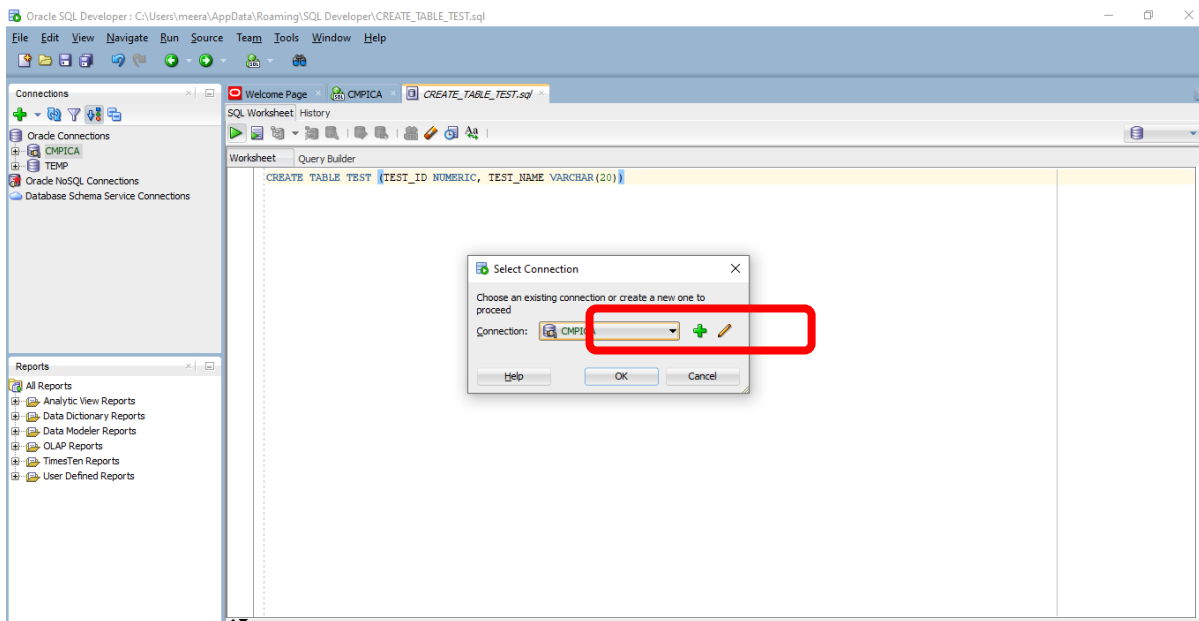
Click OK.

Step 7 Write your query and Run your query (execute) using  icon **or** press Ctrl + Enter. Select the appropriate Connection in the dialogue box and press OK.

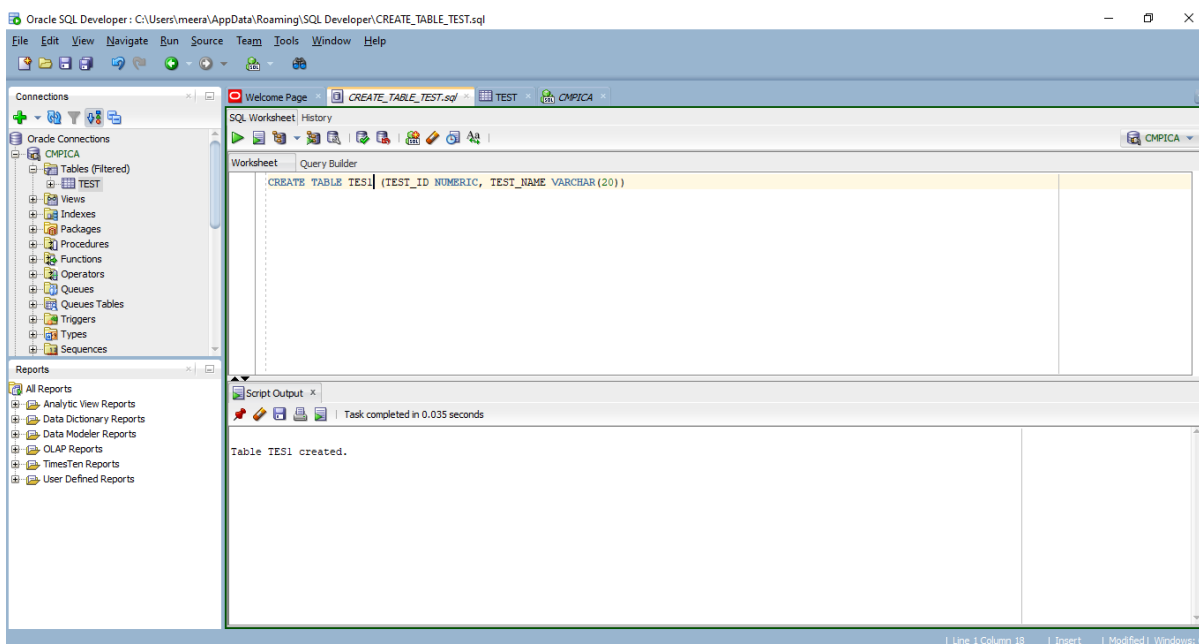


Select the appropriate connection and press OK.

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Observe execution result in bottom panel.



Congratulations!!! Your Oracle SQL Developer Setup is completed successfully.

Happy Database Exploration!!!